US012406630B2

(12) **United States Patent**
Monomoshi et al.

(10) **Patent No.: US 12,406,630 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **LUMINANCE CONTROL OF BACKLIGHT IN DISPLAY OF IMAGE**

(71) Applicant: **NICHIA CORPORATION**, Anan (JP)

(72) Inventors: **Masahiko Monomoshi**, Itano-gun (JP); **Taketoshi Nakano**, Itano-gun (JP)

(73) Assignee: **Nichia Corporation**, Anan (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/677,840**

(22) Filed: **Feb. 22, 2022**

(65) **Prior Publication Data**

US 2022/0277702 A1 Sep. 1, 2022

(30) **Foreign Application Priority Data**

Feb. 26, 2021 (JP) 2021-030121

(51) **Int. Cl.**
G09G 3/34 (2006.01)
G09G 3/36 (2006.01)

(52) **U.S. Cl.**
CPC ***G09G 3/3426*** (2013.01); ***G09G 3/3611*** (2013.01); *G09G 2320/0646* (2013.01); *G09G 2360/16* (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2010/0194791 A1* 8/2010 Ogi G06T 5/40 345/691
2011/0267379 A1 11/2011 Kurabayashi et al.
2012/0127371 A1 5/2012 Watanabe et al.
2012/0139885 A1* 6/2012 Iwasa G09G 3/3426 345/207
2012/0281028 A1* 11/2012 Orlick G09G 3/3426 345/102
2014/0055505 A1 2/2014 Ikeda et al.
2014/0240201 A1 8/2014 Takahashi et al.
(Continued)

FOREIGN PATENT DOCUMENTS

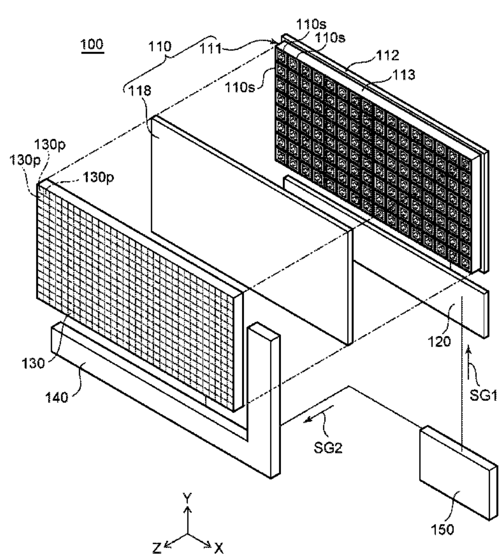
JP 2004-297439 A 10/2004
JP 2007-310037 A 11/2007
(Continued)

Primary Examiner — Benjamin C Lee
Assistant Examiner — Emily J Frank
(74) *Attorney, Agent, or Firm* — Kim & Stewart LLP

(57) **ABSTRACT**

An image display method includes generating luminance data, correcting the luminance data, generating gradation setting data, and controlling a backlight to operate based on the luminance setting data and a liquid crystal panel to operate based on the gradation setting data. The luminance data indicates a luminance value for each light-emitting region of the backlight and is generated based on a maximum gradation value among gradation values of image pixels of an input image that correspond to the light-emitting region. The luminance data is corrected such that, with respect to each light-emitting region, the luminance value is within a predetermined range below a maximum value among the luminance values of neighboring light-emitting regions, and luminance setting data is generated therefrom. The gradation setting data sets a gradation value of each pixel of the liquid crystal panel, and generated based on the input image and the luminance setting data.

20 Claims, 24 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

2016/0322005	A1	11/2016	Umeyama et al.	
2017/0084232	A1*	3/2017	Yang	G09G 3/3426
2017/0110070	A1	4/2017	Ikeda et al.	
2018/0047345	A1*	2/2018	Dunn	G02F 1/133603
2018/0166002	A1*	6/2018	Sako	G09G 3/3607
2019/0206335	A1*	7/2019	Wu	G09G 3/3426
2019/0281281	A1*	9/2019	Miyata	H04N 13/344
2020/0049877	A1	2/2020	Watanabe et al.	
2020/0105182	A1*	4/2020	Koo	G09G 3/3625
2020/0202798	A1*	6/2020	Kimbrell	G09G 3/3607
2021/0104210	A1*	4/2021	Otoi	G09G 5/10
2021/0390916	A1*	12/2021	Shi	G09G 3/342

FOREIGN PATENT DOCUMENTS

JP	2011-232590	A	11/2011
JP	2012-014628	A	1/2012
JP	2012-123100	A	6/2012
JP	2014-044302	A	3/2014
JP	2014-194525	A	10/2014
JP	2015-176137	A	10/2015
JP	2016-163099	A	9/2016
JP	2016-212195	A	12/2016
JP	2017-076110	A	4/2017
JP	2018-101521	A	6/2018
WO	2011/013745	A1	2/2011

* cited by examiner

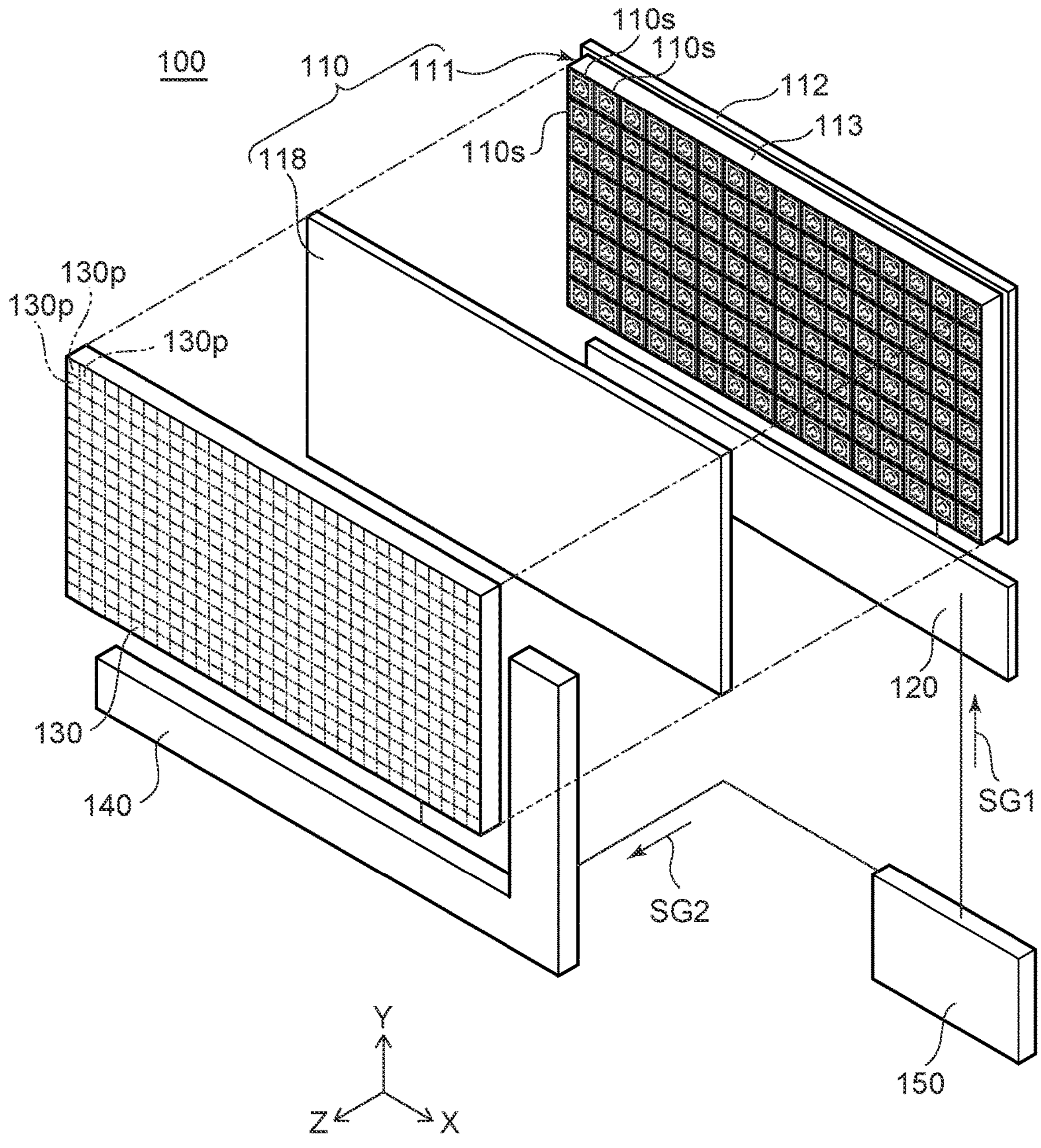

100
110
111
110s
110s
112
113
110s
118
130p
130p
130p
130
140
120
SG1
SG2
150
Y
Z
X

FIG. 1

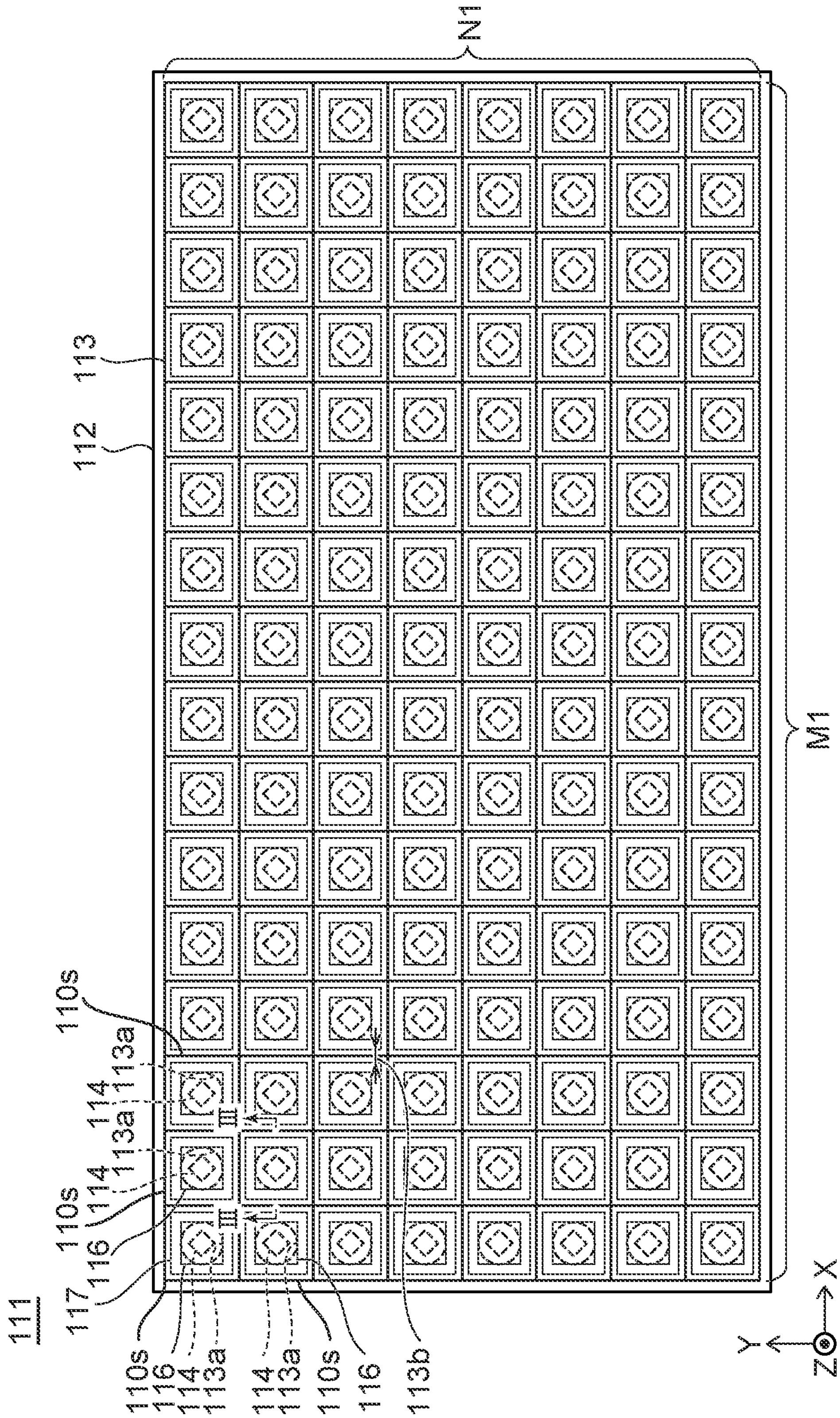


FIG. 2

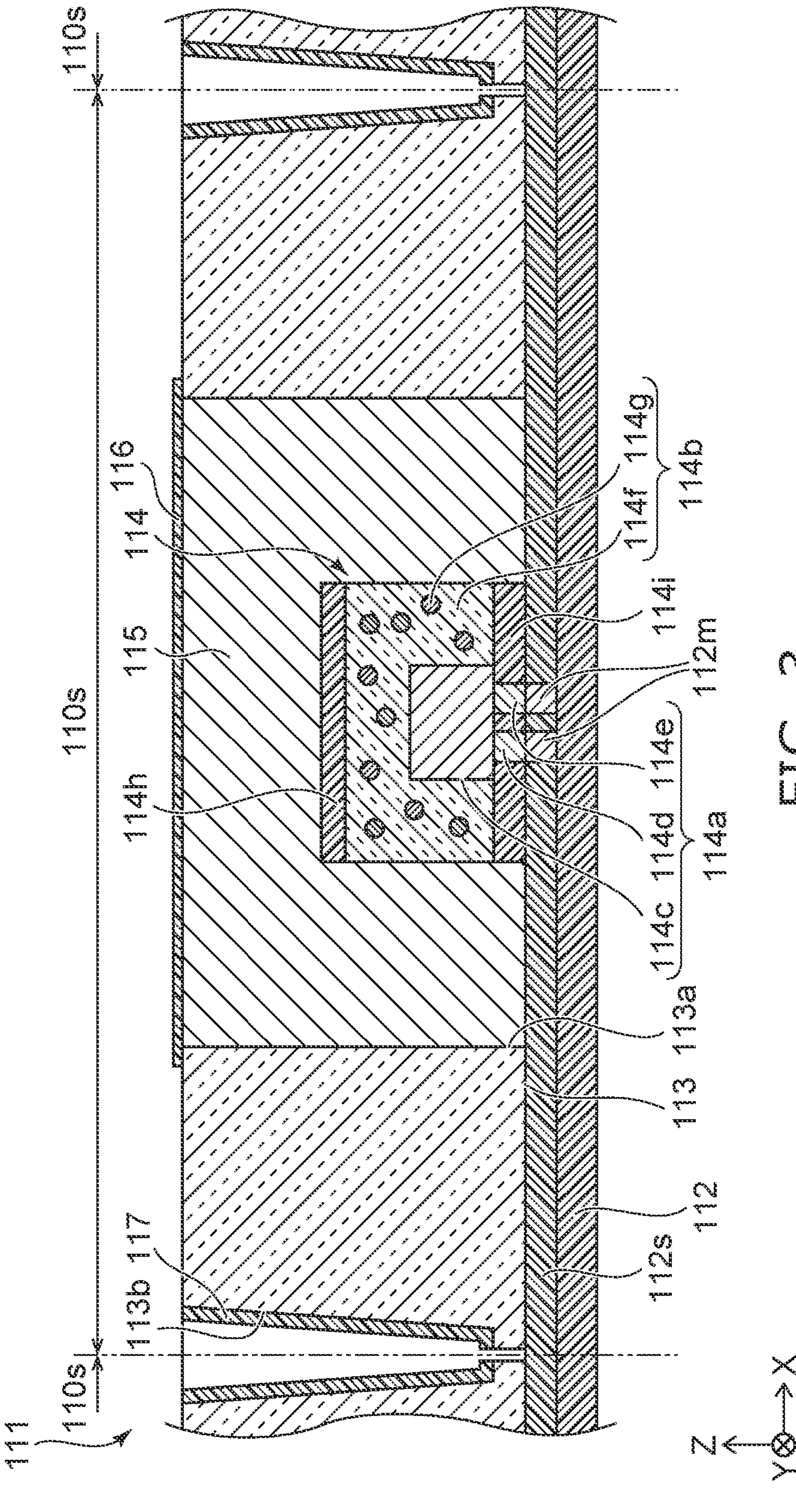

110s
110s
110s
116
114
115
114h
114g
114f
114b
114i
112m
114e
114d
114c
114a
113a
113
112
112s
113b
117
111
X
Y
Z

FIG. 3

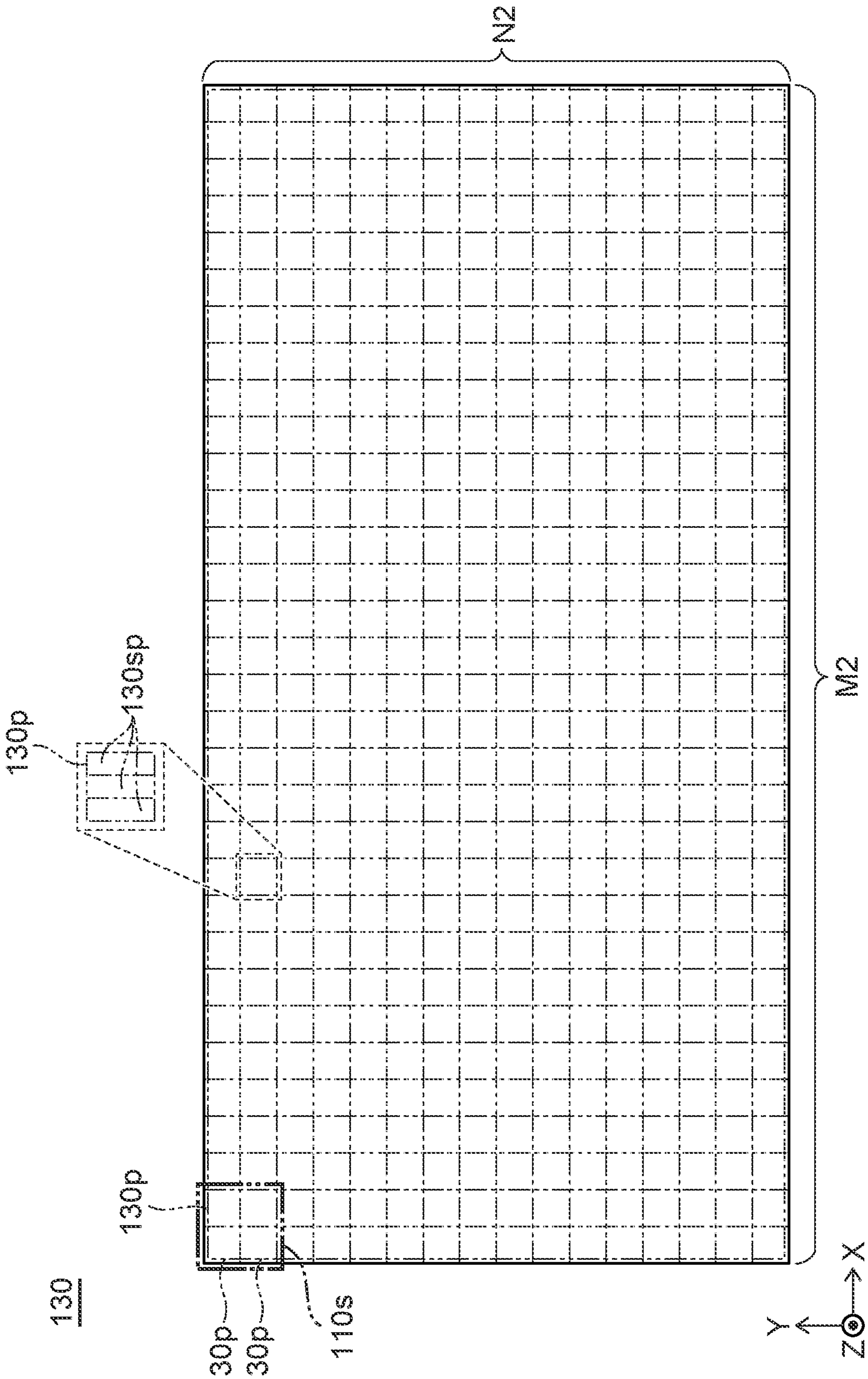


FIG. 4

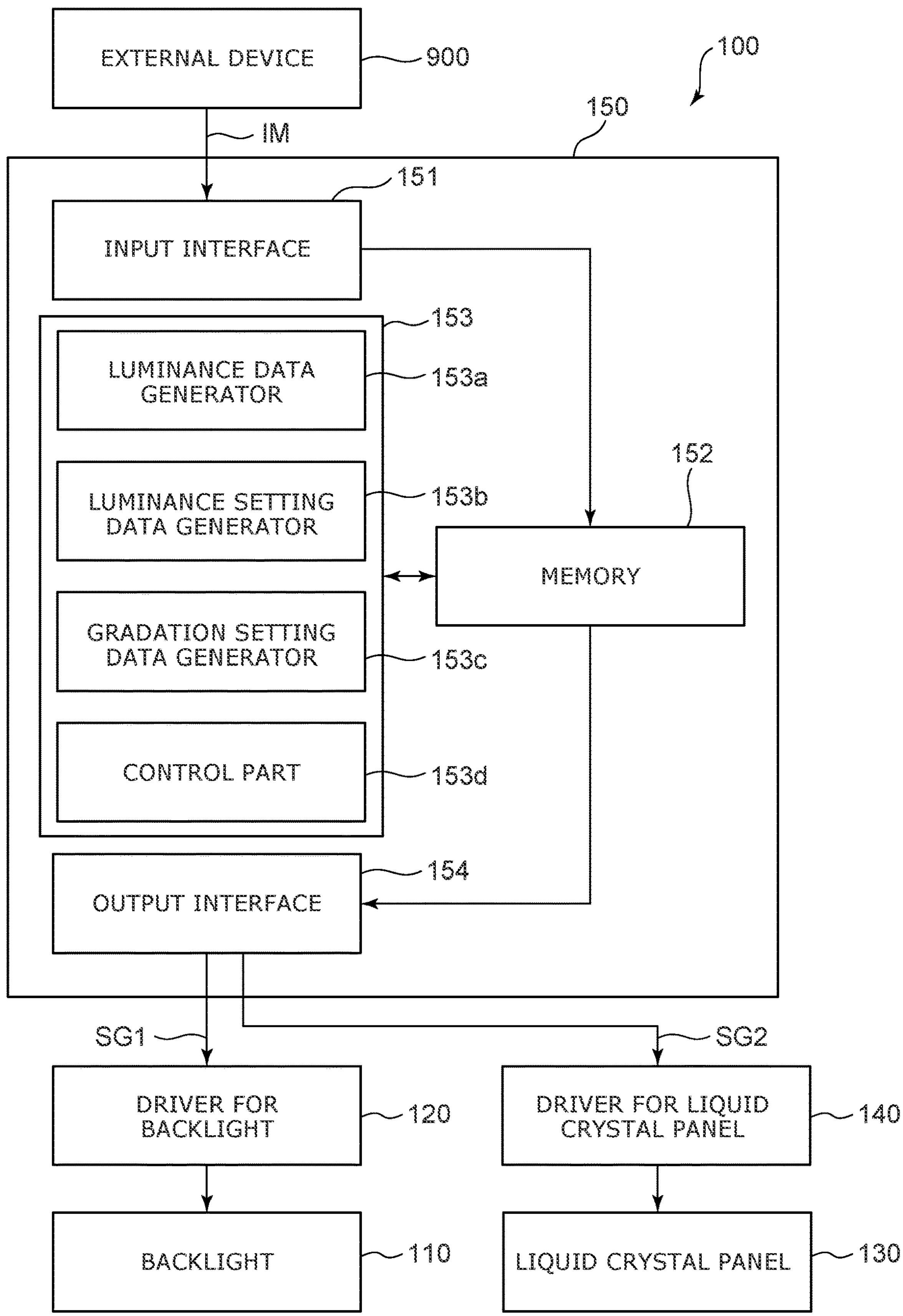

EXTERNAL DEVICE
900
100
IM
150
151
INPUT INTERFACE
153
LUMINANCE DATA GENERATOR
153a
152
LUMINANCE SETTING DATA GENERATOR
153b
MEMORY
GRADATION SETTING DATA GENERATOR
153c
CONTROL PART
153d
154
OUTPUT INTERFACE
SG1
SG2
DRIVER FOR BACKLIGHT
120
DRIVER FOR LIQUID CRYSTAL PANEL
140
BACKLIGHT
110
LIQUID CRYSTAL PANEL
130

FIG. 5

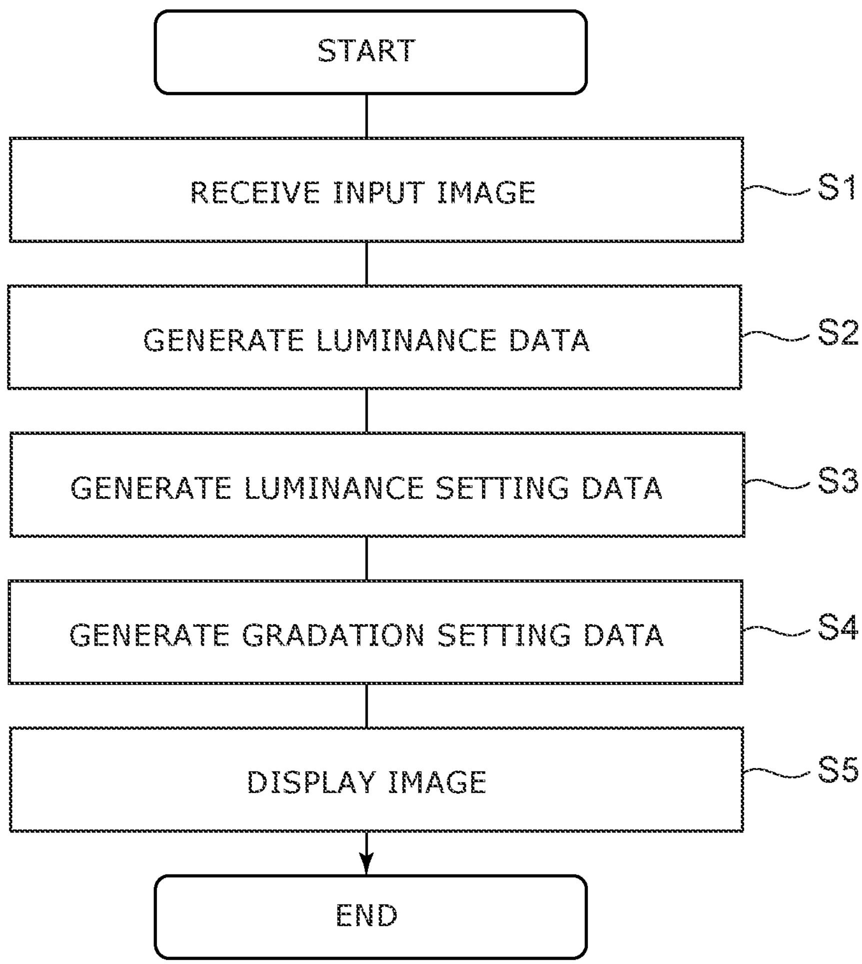

START
RECEIVE INPUT IMAGE
S1
GENERATE LUMINANCE DATA
S2
GENERATE LUMINANCE SETTING DATA
S3
GENERATE GRADATION SETTING DATA
S4
DISPLAY IMAGE
S5
END

FIG. 6

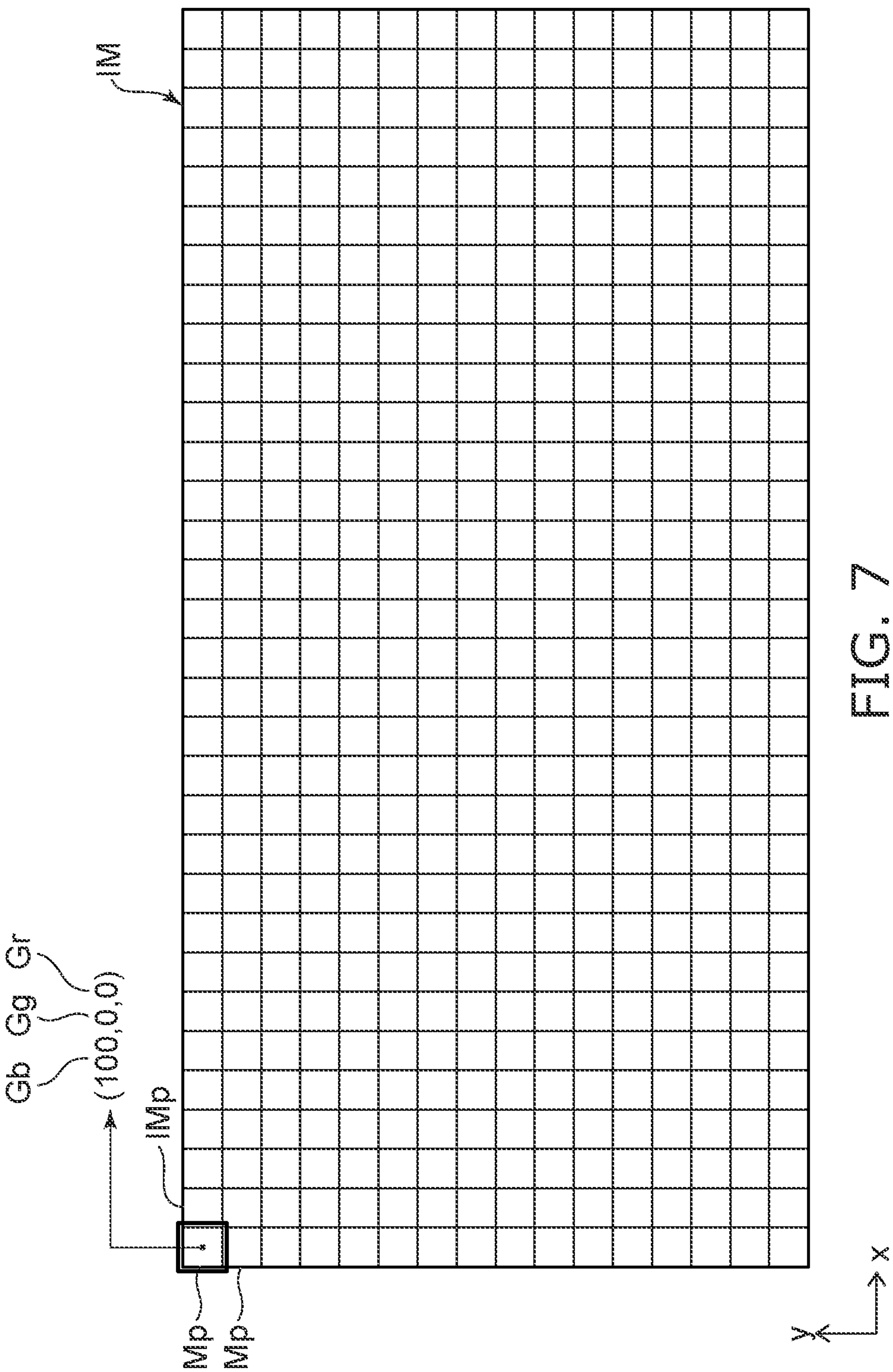


FIG. 7

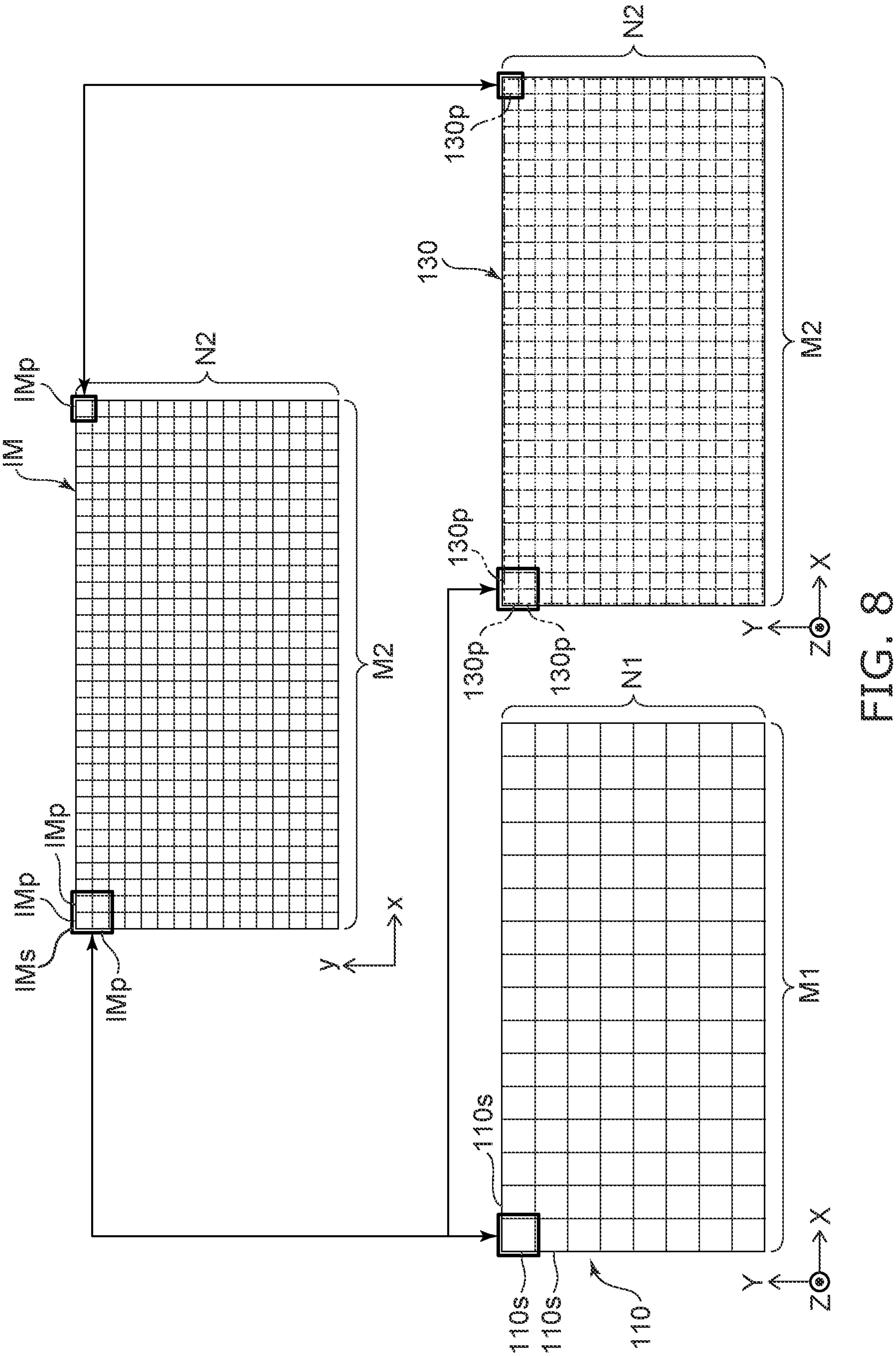


FIG. 8

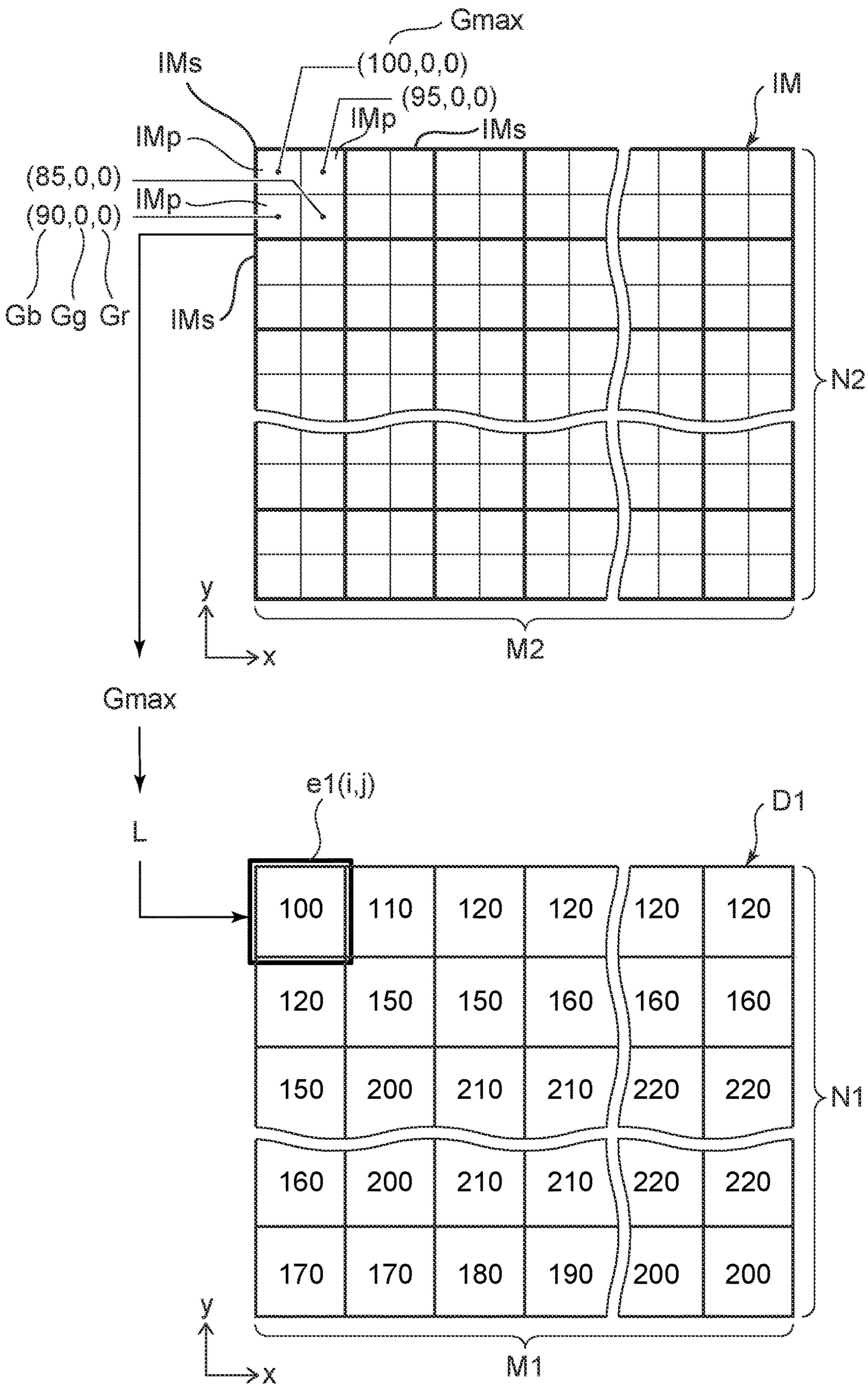

Gmax
IMs
(100,0,0)
(95,0,0)
IMp
IMs
IM
IMp
(85,0,0)
IMp
(90,0,0)
Gb Gg Gr
IMs
N2
y
x
M2
Gmax
L
e1(i,j)
D1
100 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
N1
y
x
M1

FIG. 9

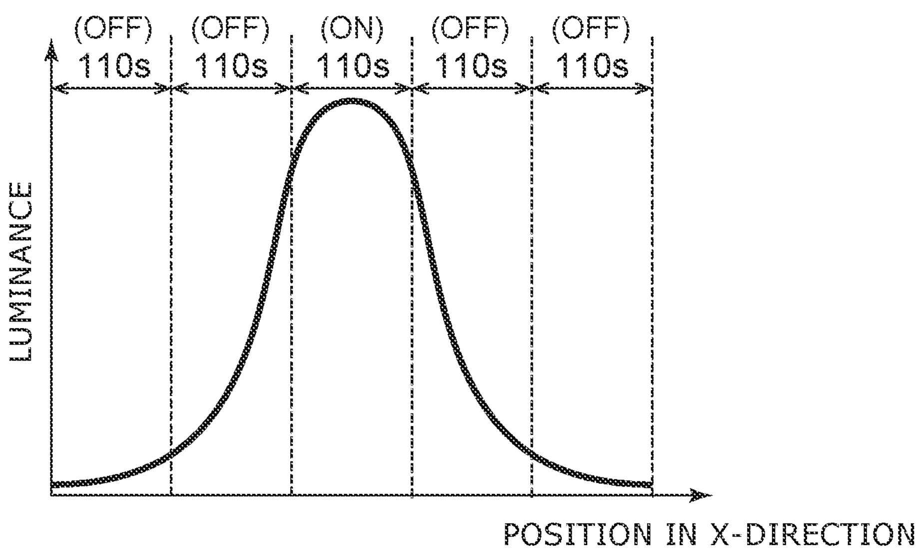

(OFF)
110s
(OFF)
110s
(ON)
110s
(OFF)
110s
(OFF)
110s
LUMINANCE
POSITION IN X-DIRECTION

FIG. 10

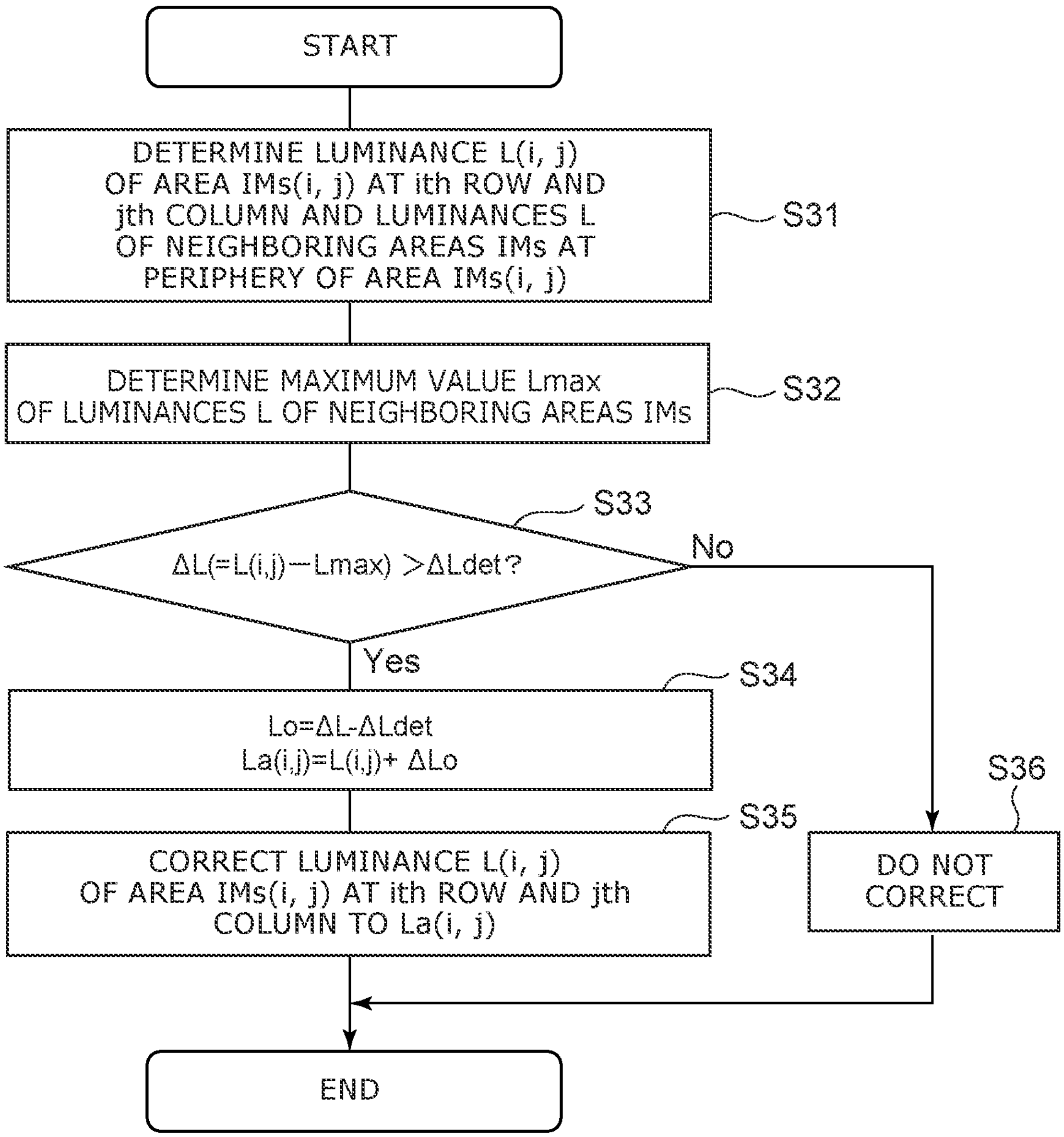

START
DETERMINE LUMINANCE L(i, j) OF AREA IMs(i, j) AT ith ROW AND jth COLUMN AND LUMINANCES L OF NEIGHBORING AREAS IMs AT PERIPHERY OF AREA IMs(i, j)
S31
DETERMINE MAXIMUM VALUE Lmax OF LUMINANCES L OF NEIGHBORING AREAS IMs
S32
S33
ΔL(=L(i,j)−Lmax) >ΔLdet?
No
Yes
S34
Lo=ΔL-ΔLdet
La(i,j)=L(i,j)+ ΔLo
S35
CORRECT LUMINANCE L(i, j) OF AREA IMs(i, j) AT ith ROW AND jth COLUMN TO La(i, j)
S36
DO NOT CORRECT
END

FIG. 11

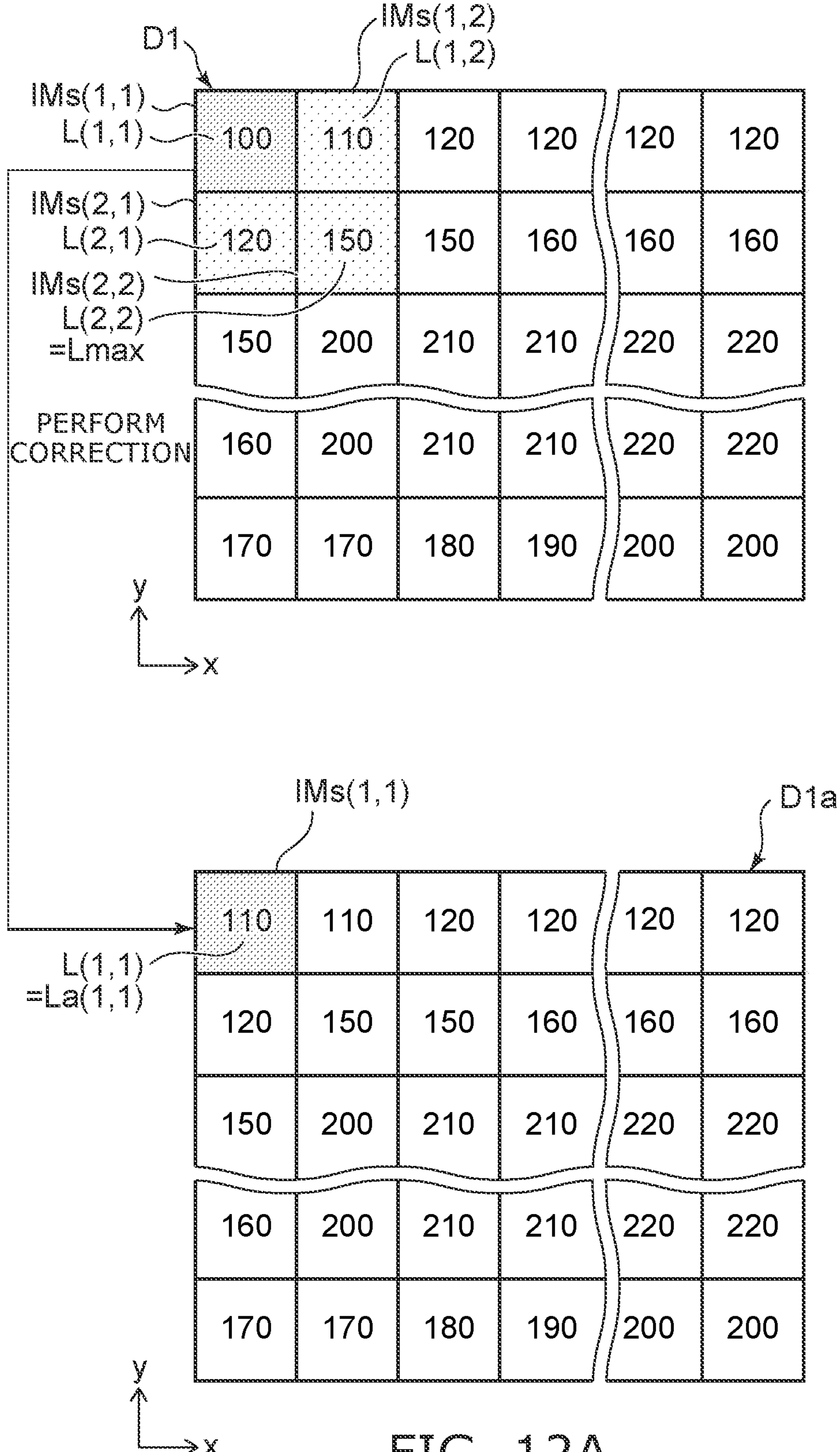

D1
IMs(1,2)
L(1,2)
IMs(1,1)
L(1,1)
IMs(2,1)
L(2,1)
IMs(2,2)
L(2,2)
=Lmax
PERFORM
CORRECTION
100 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
y
x
IMs(1,1)
D1a
L(1,1)
=La(1,1)
110 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
y
x

FIG. 12A

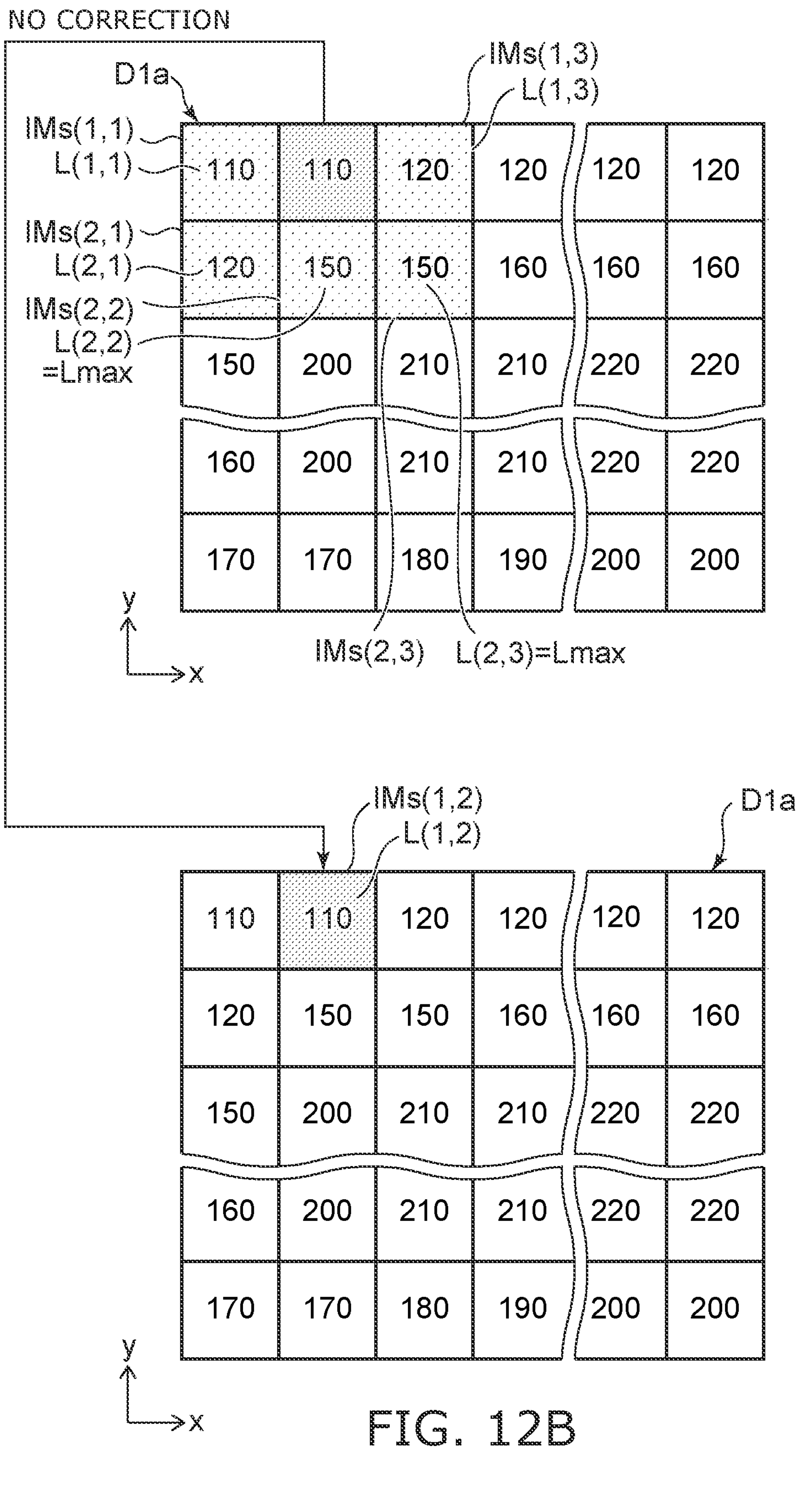

NO CORRECTION
D1a
IMs(1,1)
L(1,1)
IMs(2,1)
L(2,1)
IMs(2,2)
L(2,2)
=Lmax
IMs(1,3)
L(1,3)
IMs(2,3)
L(2,3)=Lmax
110 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
y
x
IMs(1,2)
L(1,2)
D1a
110 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
y
x

FIG. 12B

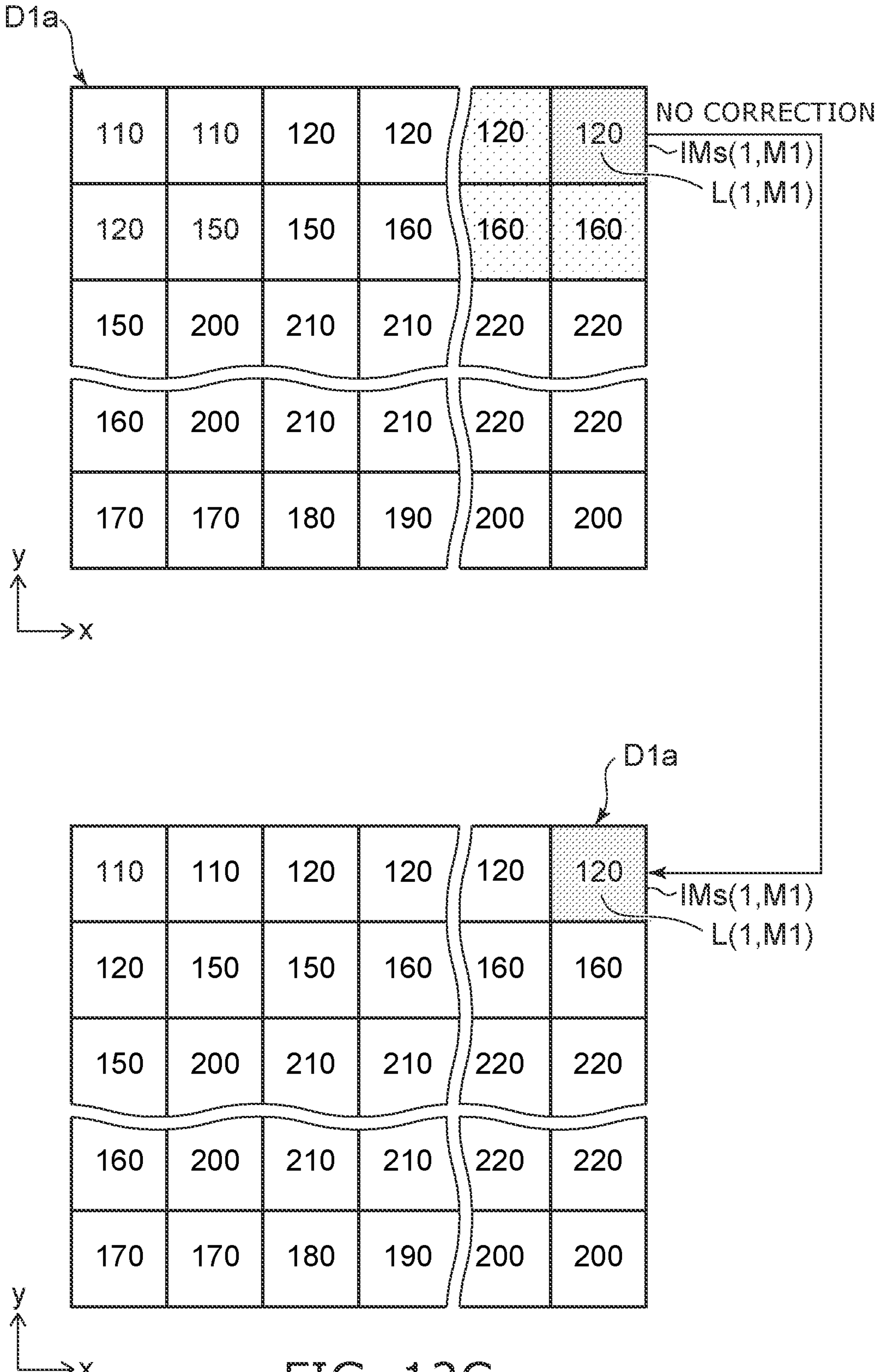

D1a
110 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
NO CORRECTION
IMs(1,M1)
L(1,M1)
y
x
D1a
110 110 120 120 120 120
120 150 150 160 160 160
150 200 210 210 220 220
160 200 210 210 220 220
170 170 180 190 200 200
IMs(1,M1)
L(1,M1)
y
x

FIG. 12C

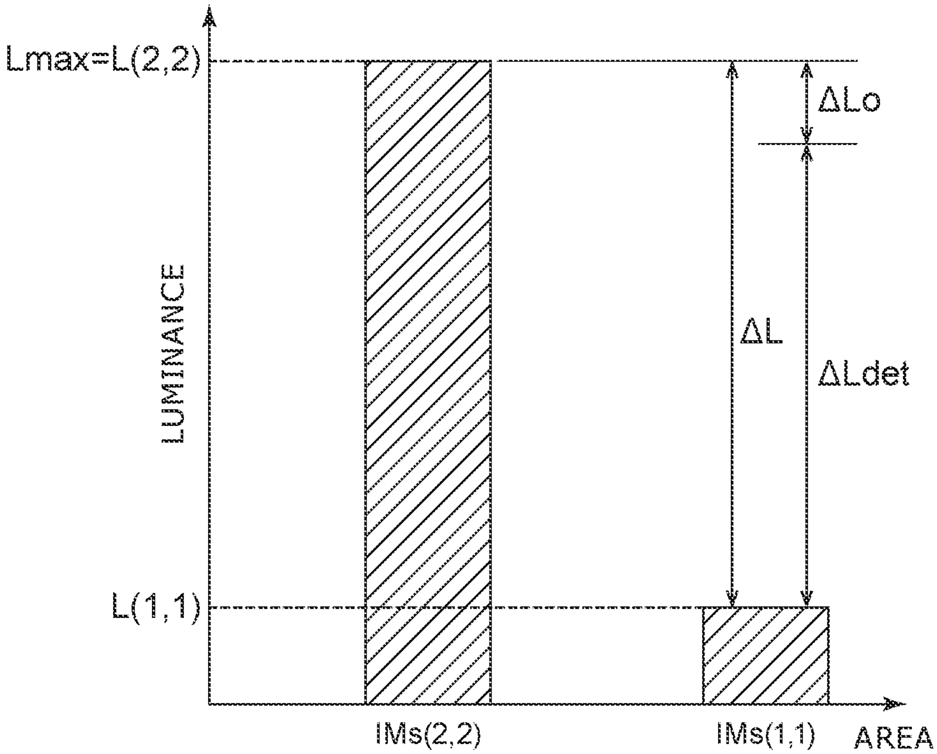


FIG. 13A

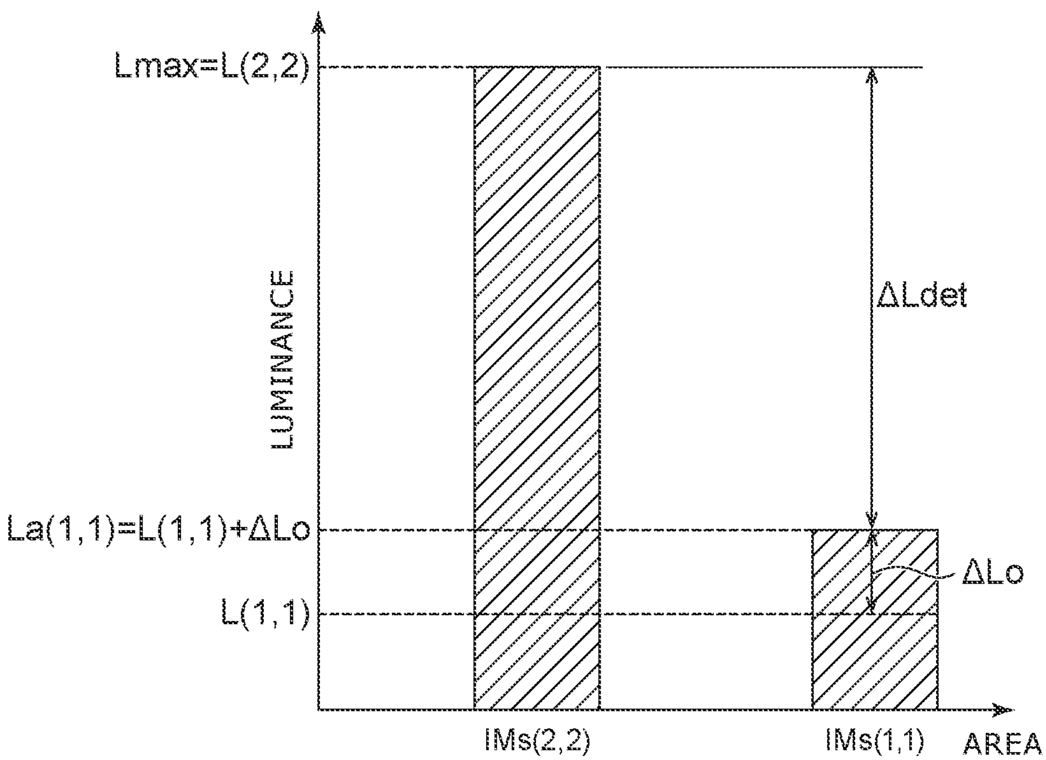


FIG. 13B

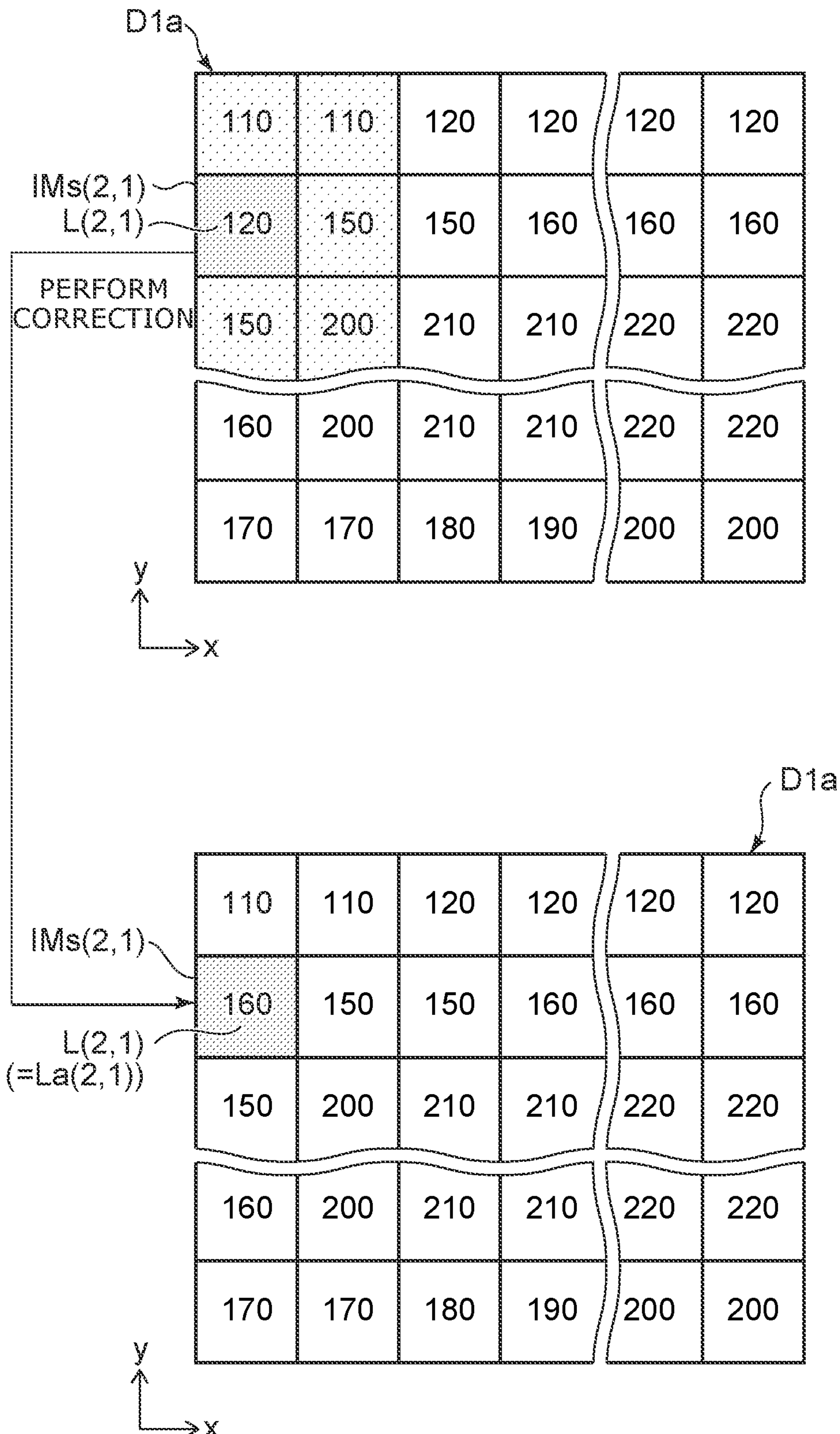

D1a
IMs(2,1)
L(2,1)
PERFORM CORRECTION
110
110
120
120
120
120
120
150
150
160
160
160
150
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
y
x
D1a
IMs(2,1)
L(2,1)
(=La(2,1))
110
110
120
120
120
120
160
150
150
160
160
160
150
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
y
x

FIG. 14A

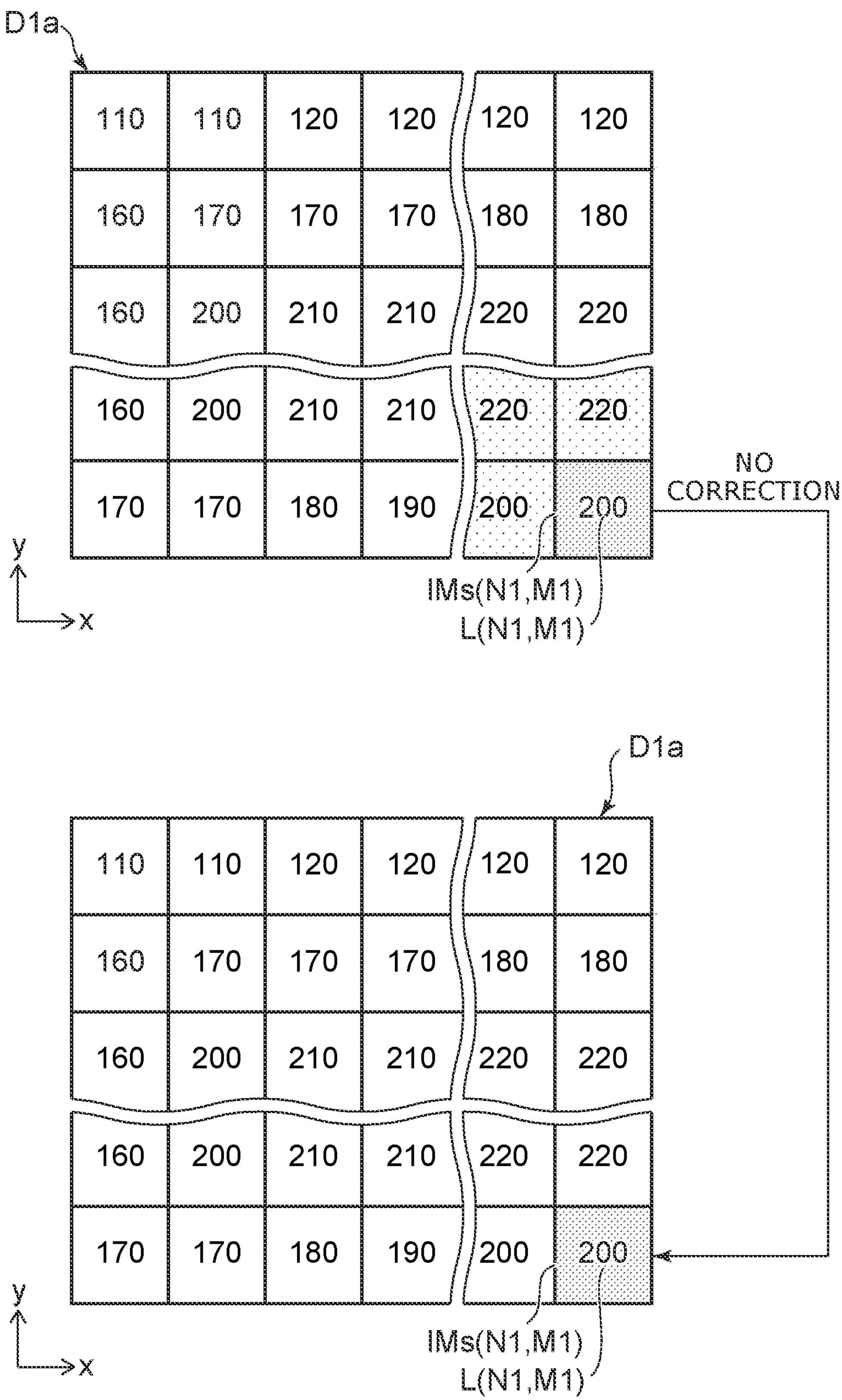

D1a
110
110
120
120
120
120
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
NO CORRECTION
IMs(N1,M1)
L(N1,M1)
y
x
D1a
110
110
120
120
120
120
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
IMs(N1,M1)
L(N1,M1)
y
x

FIG. 14B

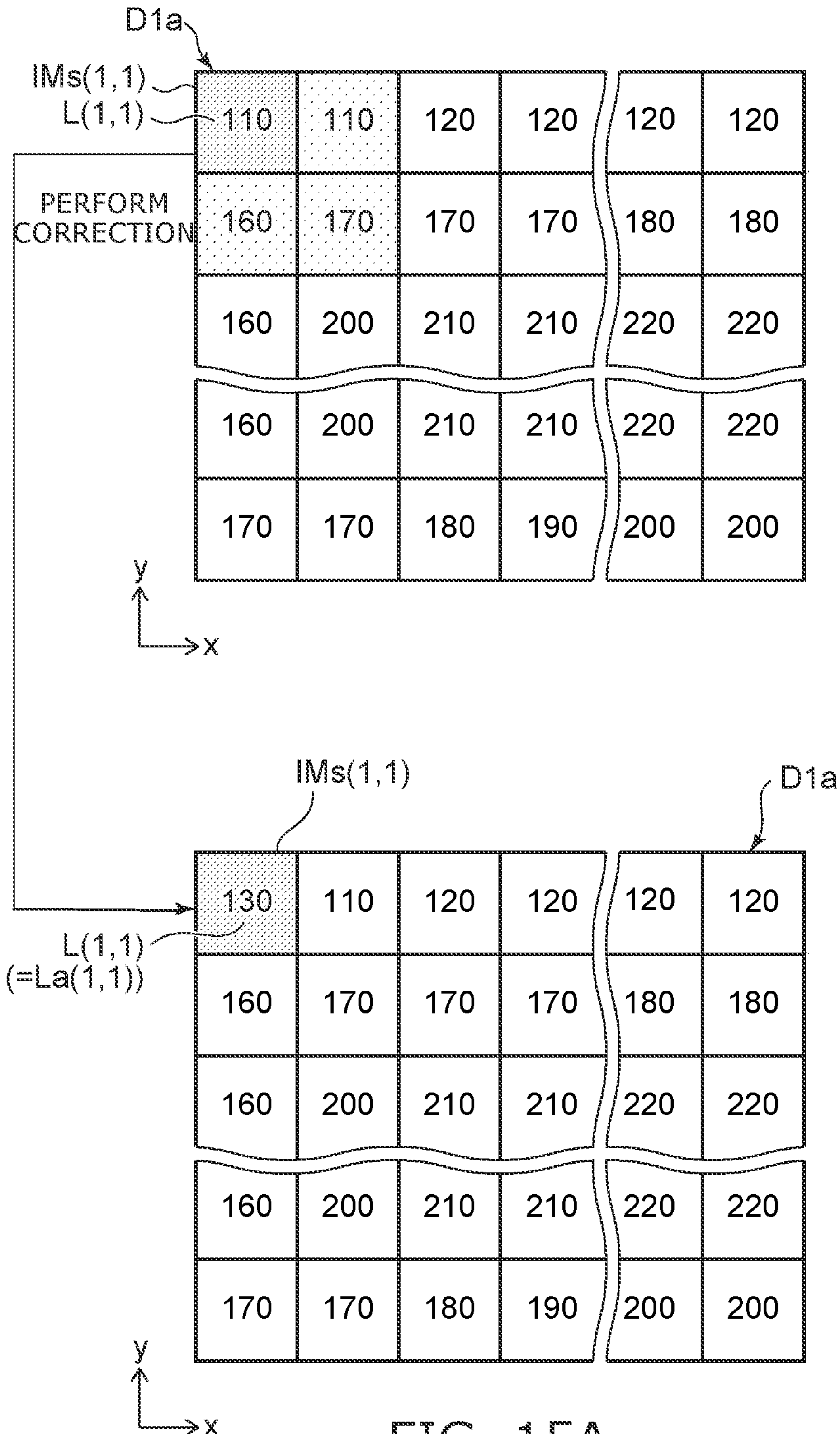

D1a
IMs(1,1)
L(1,1)
110
110
120
120
120
120
PERFORM CORRECTION
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
y
x
IMs(1,1)
D1a
130
110
120
120
120
120
L(1,1)
(=La(1,1))
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
y
x

FIG. 15A

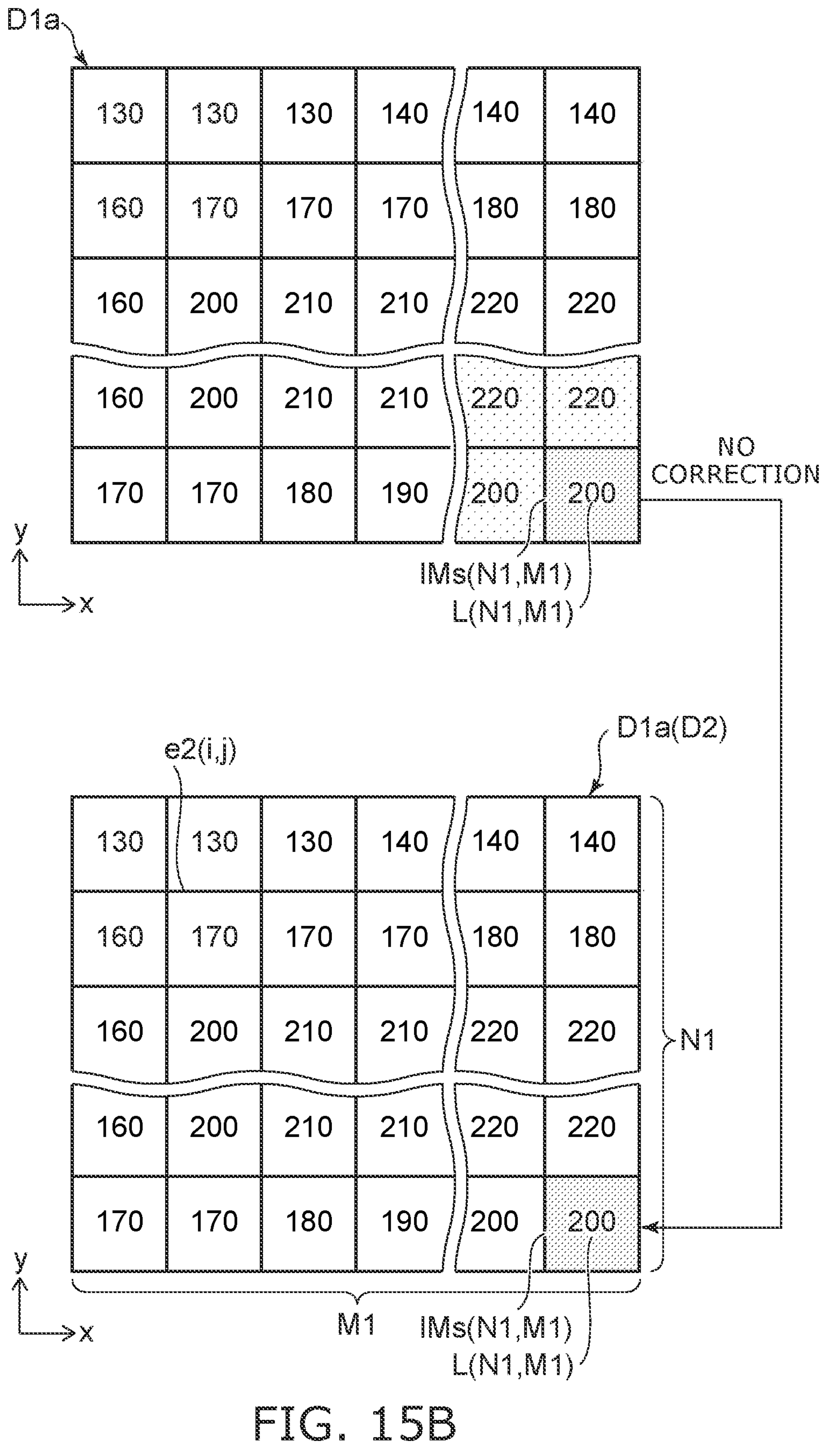

D1a
130
130
130
140
140
140
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
NO CORRECTION
IMs(N1,M1)
L(N1,M1)
y
x
D1a(D2)
e2(i,j)
130
130
130
140
140
140
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
N1
M1
IMs(N1,M1)
L(N1,M1)
y
x

FIG. 15B

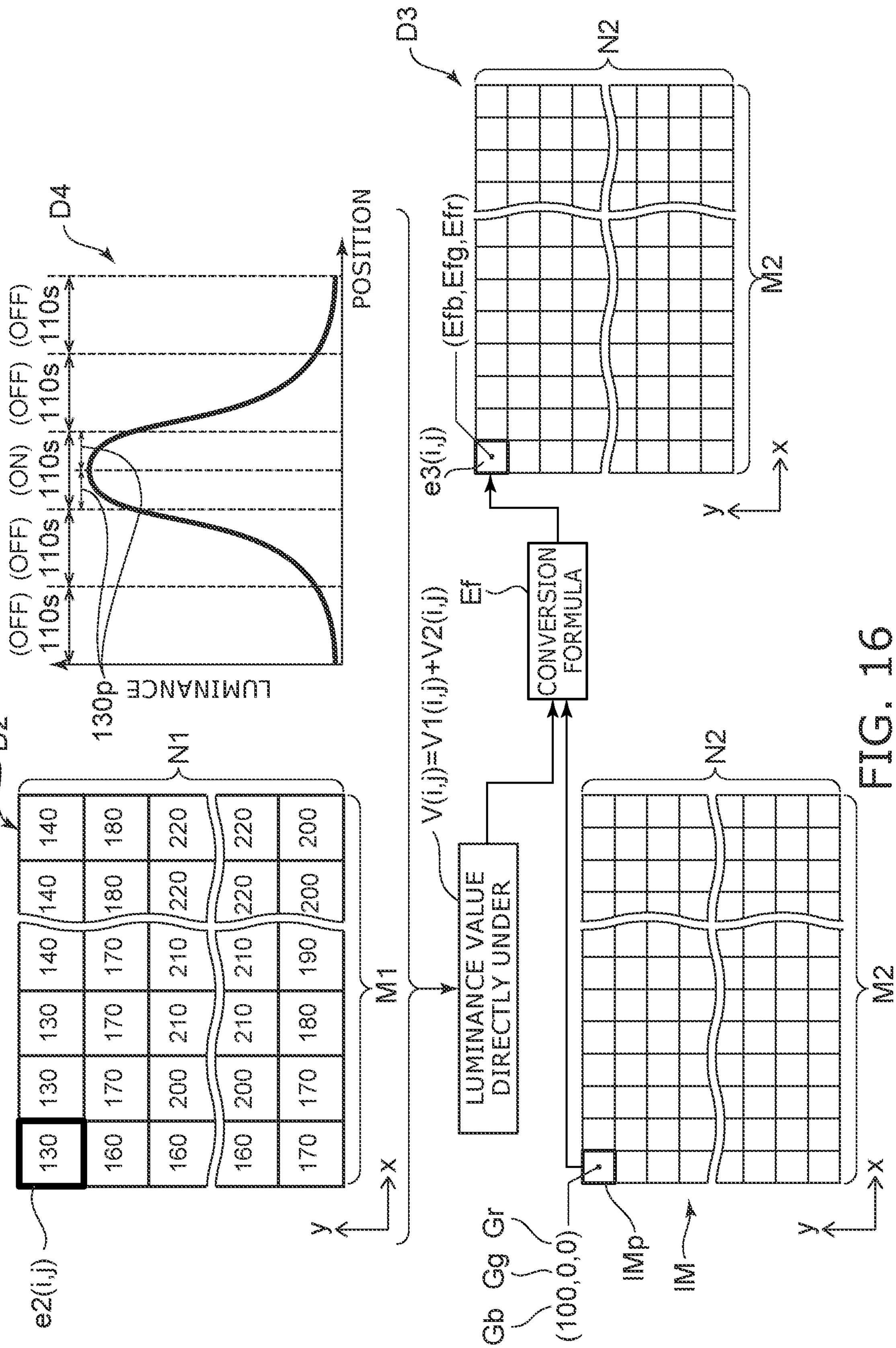

D2
e2(i,j)
130
130
130
140
140
140
160
170
170
170
180
180
160
200
210
210
220
220
160
200
210
210
220
220
170
170
180
190
200
200
N1
M1
x
y
D4
(OFF)
(OFF)
(OFF)
(ON)
(OFF)
(OFF)
(OFF)
110s
130p
LUMINANCE
POSITION
V(i,j)=V1(i,j)+V2(i,j)
LUMINANCE VALUE
DIRECTLY UNDER
Ef
CONVERSION
FORMULA
D3
N2
M2
e3(i,j)
(Efb,Efg,Efr)
Gb Gg Gr
(100,0,0)
IMp
IM

FIG. 16

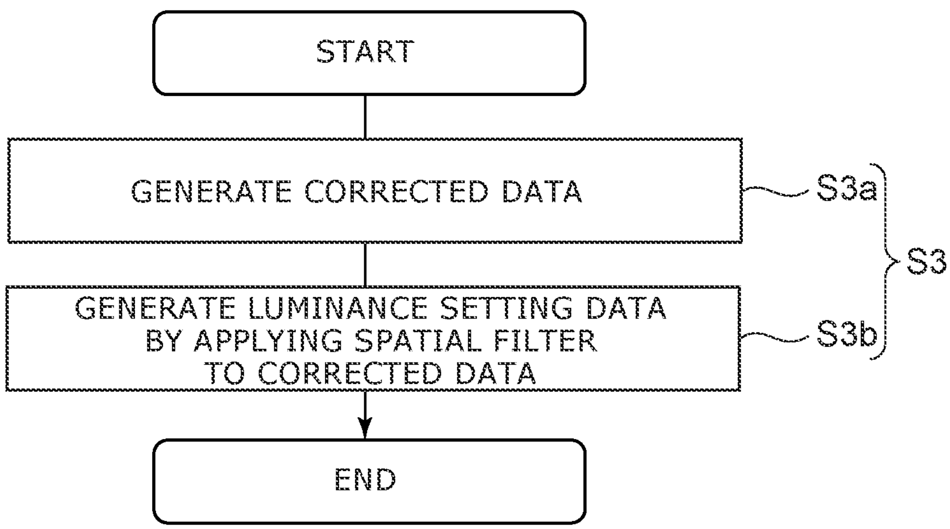

START
GENERATE CORRECTED DATA
S3a
S3
GENERATE LUMINANCE SETTING DATA
BY APPLYING SPATIAL FILTER
TO CORRECTED DATA
S3b
END

FIG. 17

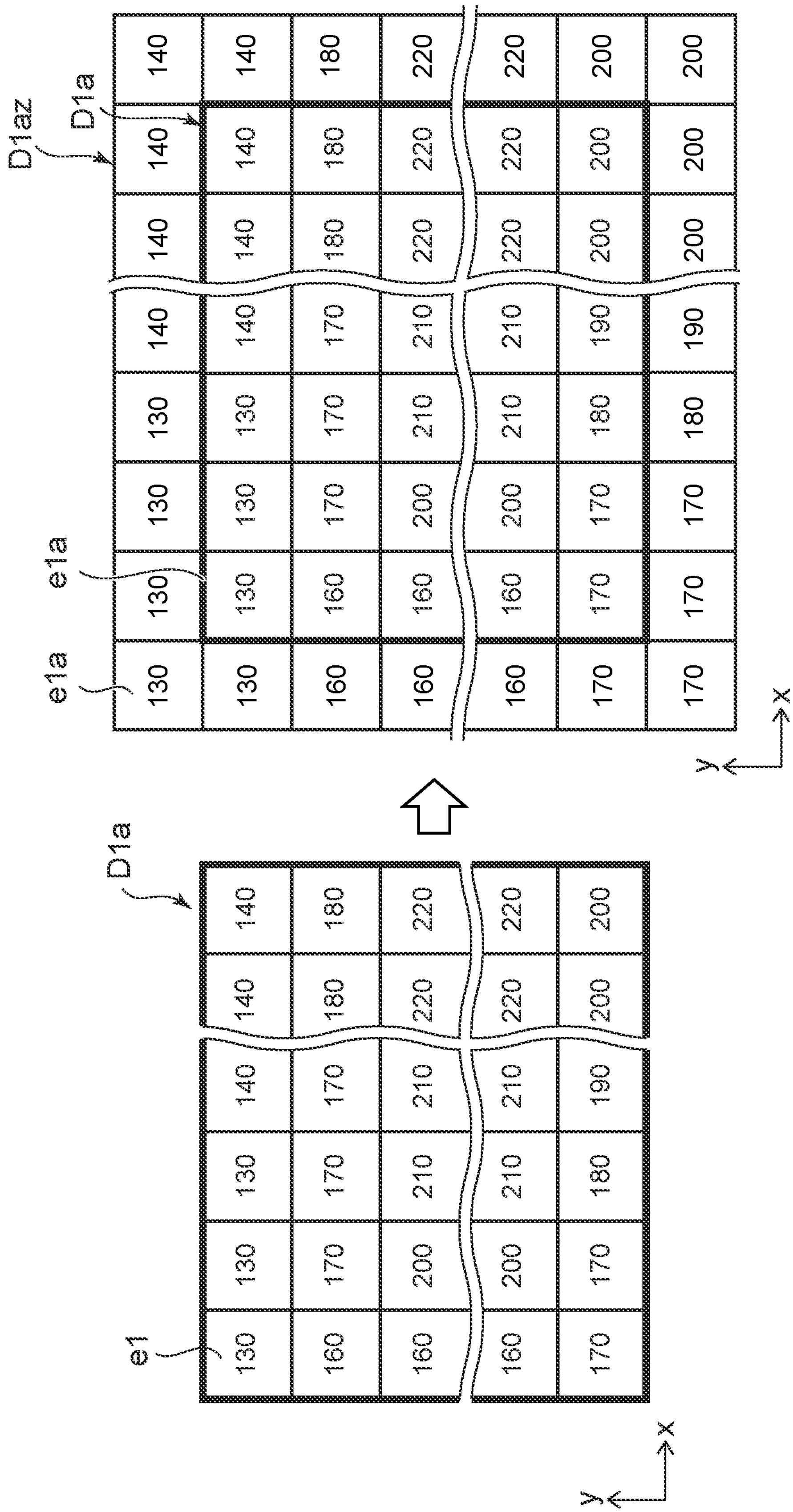


FIG. 18

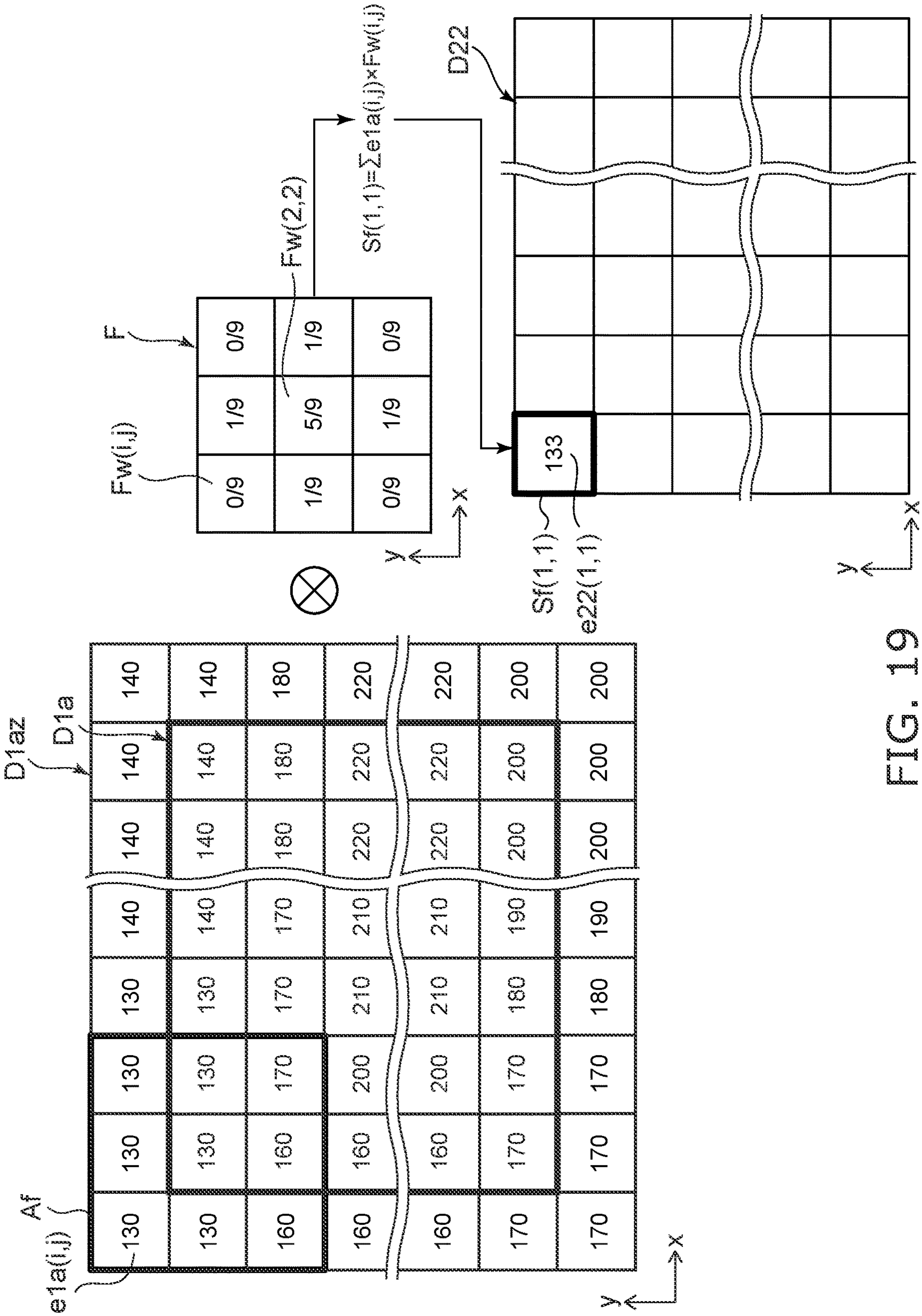


FIG. 19

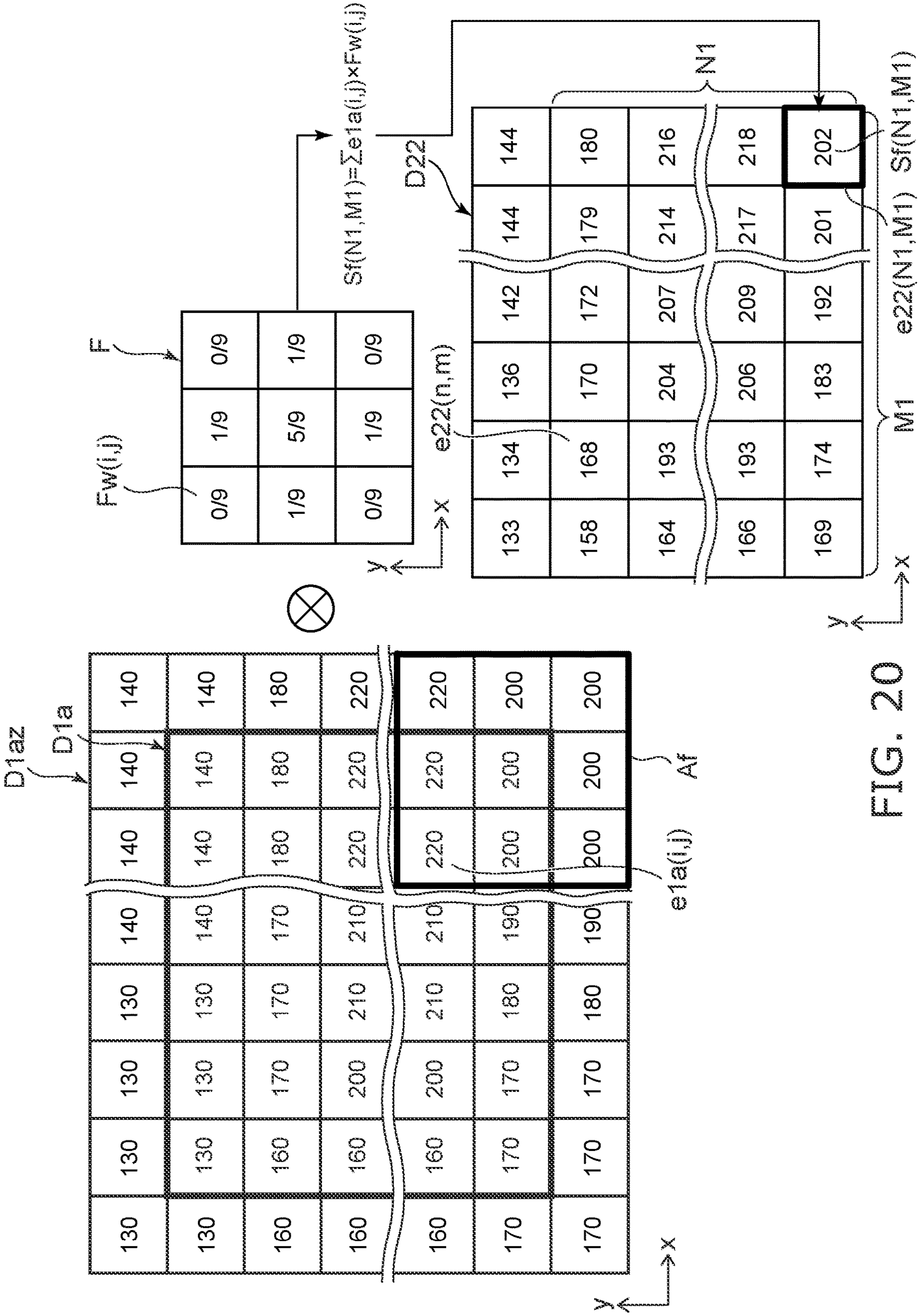

D1az
D1a
Af
e1a(i,j)
F
Fw(i,j)
Sf(N1,M1)=Σe1a(i,j)×Fw(i,j)
D22
e22(n,m)
N1
M1
e22(N1,M1)
Sf(N1,M1)

FIG. 20

LUMINANCE CONTROL OF BACKLIGHT IN DISPLAY OF IMAGE

CROSS-REFERENCE TO RELATED APPLICATION

This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-030121, filed on Feb. 26, 2021; the entire contents of which are incorporated herein by reference.

FIELD

Embodiments relate to an image display method and a display that performs the same.

BACKGROUND

A conventionally-known image display device includes a backlight and a liquid crystal panel. The backlight includes multiple light-emitting regions arranged in a matrix configuration and light sources in the light-emitting regions. The liquid crystal panel is located above the backlight and includes multiple pixels. By using such an image display device, luminances of the light-emitting regions can be set differently depending on an image to be displayed on the liquid crystal panel. Also, gradations of the pixels of the liquid crystal panel can be set according to the set luminances of the light-emitting regions. The contrast of the image can be improved thereby. Such technology is called "local dimming".

A backlight that is used for the local dimming may have a structure in which light can propagate (i.e., leak) between the adjacent light-emitting regions. When a backlight having such a structure is used for the local dimming, the leakage of the light becomes more significant and thus noticeable by users as a difference between setting values of luminances of the adjacent light-emitting regions increases. Such a phenomenon is called a "halo phenomenon".

SUMMARY

Embodiments are directed to an image display method and a display in which the halo phenomenon can be suppressed.

An image display method includes generating luminance data, performing correction of the luminance data, generating gradation setting data, and controlling a backlight to operate based on the luminance setting data and a liquid crystal panel to operate based on the gradation setting data to display an image corresponding to an input image. The luminance data indicates a luminance value for each of a plurality of light-emitting regions of the backlight configured in a matrix form and is generated based on a maximum gradation value among gradation values of image pixels of the input image that correspond to the light-emitting region. The luminance data is corrected such that, with respect to each of the light-emitting regions, the luminance value is within a predetermined range below a maximum luminance value among the luminance values of neighboring light-emitting regions thereof, and luminance setting data is generated therefrom. The gradation setting data sets a gradation value of each of the pixels of the liquid crystal panel for the input image, and generated based on the input image and the luminance setting data.

According to embodiments, the halo phenomenon can be suppressed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates an exploded perspective view of an image display device according to a first embodiment;

FIG. **2** illustrates a top view of a planar light source of a backlight included in the image display device according to the first embodiment;

FIG. **3** illustrates a cross-sectional view of the planar light source along line III-III in FIG. **2**;

FIG. **4** illustrates a top view of a liquid crystal panel of the image display device according to the first embodiment;

FIG. **5** is a block diagram showing components of the image display device according to the first embodiment;

FIG. **6** is a flowchart showing an image display method according to the first embodiment;

FIG. **7** is a schematic diagram showing an input image input to a controller of the image display device according to the first embodiment;

FIG. **8** is a schematic diagram showing a relationship among pixels of the liquid crystal panel, light-emitting regions of the backlight, and pixels of the input image in the first embodiment;

FIG. **9** is a schematic diagram showing a process of generating luminance data in the image display method according to the first embodiment;

FIG. **10** is a graph showing a luminance distribution when a light source in one light-emitting region is lit in the backlight of the image display device according to the first embodiment;

FIG. **11** is a flowchart showing a process of correcting the luminance of one area of the luminance data in the image display method according to the first embodiment;

FIGS. **12**A, **12**B, and **12**C are schematic diagrams showing a part of the process of generating luminance setting data in the image display method according to the first embodiment;

FIGS. **13**A and **13**B are schematic diagrams showing luminance adjustment in the process of generating the luminance setting data in the image display method according to the first embodiment;

FIGS. **14**A and **14**B are schematic diagrams showing another part of the process of generating the luminance setting data in the image display method according to the first embodiment;

FIGS. **15**A and **15**B are schematic diagrams showing another part of the process of generating the luminance setting data in the image display method according to the first embodiment;

FIG. **16** is a schematic diagram showing a process of generating gradation setting data in the image display method according to the first embodiment;

FIG. **17** is a flowchart showing a process of generating luminance setting data in an image display method according to a second embodiment;

FIG. **18** is a schematic diagram a part of a process of generating the luminance setting data in the image display method according to the second embodiment;

FIG. **19** is a schematic diagram showing another part of the process of generating the luminance setting data in the image display method according to the second embodiment; and

FIG. **20** is a schematic diagram showing another part of the process of generating the luminance setting data in the image display method according to the second embodiment.

DETAILED DESCRIPTION

Exemplary embodiments will now be described with reference to the drawings. The drawings are schematic or conceptual; and the relationships between the thickness and width of portions, the proportional coefficients of sizes among portions, etc., are not necessarily the same as actual values thereof. Furthermore, the dimensions and proportional coefficients may be illustrated differently among the drawings, even for identical portions. In the specification and the drawings of the application, components similar to those described in regard to a drawing hereinabove are marked with like reference numerals, and a detailed description is omitted as appropriate.

For easier understanding of the following description, arrangements and configurations of portions of an image display device are described using an XYZ orthogonal coordinate system. X-axis, Y-axis, and Z-axis are orthogonal to each other. The direction in which the X-axis extends is referred to as an "X-direction"; the direction in which the Y-axis extends is referred to as a "Y-direction"; and the direction in which the Z-axis extends is referred to as a "Z-direction". For easier understanding of the description, the Z-direction is called up, and the opposite direction is called down, but these directions are independent of the direction of gravity. For easier understanding of the description of the drawings, the X-axis direction in the direction of the arrow is referred to as the "+X direction"; and the opposite direction is referred to as the "−X direction". Similarly, the Y-axis direction in the direction of the arrow is referred to as the "+Y direction"; and the opposite direction is referred to as the "−Y direction".

First Embodiment

First, a first embodiment will be described.

FIG. **1** illustrates an exploded perspective view of an image display device according to the first embodiment.

An image display device **100** according to the first embodiment is, for example, a liquid crystal module (LCM) used in a display of a device such as a television, a personal computer, a game machine, etc. The image display device **100** includes a backlight **110**, a driver **120** for the backlight, a liquid crystal panel **130**, a driver **140** for the liquid crystal panel, and a controller **150**. Components of the image display device **100** will be described hereinafter. For easier understanding of the description, the electrical connections between the components are shown by connecting the components to each other with solid lines in FIG. **1**.

The backlight **110** is compatible with local dimming. The backlight **110** includes a planar light source **111**, and an optical member **118** located on the planar light source **111**.

Although not particularly limited, the optical member **118** is, for example, a sheet, a film, or a plate that has a light-modulating function such as a light-diffusing function, etc. According to the present embodiment, the number of the optical members **118** included in the backlight **110** is one. However, the number of optical members included in the backlight may be two or more.

FIG. **2** illustrates a top view of the planar light source **111** of the backlight **110** included in the image display device **100** according to the first embodiment.

FIG. **3** illustrates a cross-sectional view of the planar light source **111** along line III-III in FIG. **2**.

According to the first embodiment as shown in FIGS. **2** and **3**, the planar light source **111** includes a substrate **112**, a light-reflective sheet **112*s***, a light guide member **113**, multiple light sources **114**, a light-transmitting member **115**, a first light-modulating member **116**, and a light-reflecting member **117**.

The substrate **112** is a wiring substrate that includes an insulating member, and multiple wiring located in the insulating member. According to the present embodiment, the shape of the substrate **112** in top-view is substantially rectangular as shown in FIG. **2**. However, the shape of the substrate is not limited to the aforementioned shape. The upper surface and the lower surface of the substrate **112** are flat surfaces and are substantially parallel to the X-direction and the Y-direction.

As shown in FIG. **3**, the light-reflective sheet **112*s*** is located on the substrate **112**. According to the present embodiment, the light-reflective sheet **112*s*** includes a first adhesive layer, a light-reflecting layer on the first adhesive layer, and a second adhesive layer on the light-reflecting layer. The light-reflective sheet **112*s*** is adhered to the substrate **112** with the first adhesive layer.

The light guide member **113** is located on the light-reflective sheet **112*s***. At least a portion of the lower surface of the light guide member **113** is adhered to the light-reflective sheet **112*s*** with the second adhesive layer. According to the present embodiment, the light guide member **113** is plate-shaped. The thickness of the light guide member **113** is preferably, for example, not less than 200 μm and not more than 800 μm. In the thickness direction, the light guide member **113** may include a single layer or may include a stacked body of multiple layers. According to the present embodiment, the shape of the light guide member **113** in top-view is substantially rectangular as shown in FIG. **2**. However, the shape of the light guide member is not limited to the aforementioned shape.

For example, a thermoplastic resin such as acrylic, polycarbonate, cyclic polyolefin, polyethylene terephthalate, polyester, or the like, an epoxy, a thermosetting resin such as silicone or the like, and glass, etc., can be used as a material used for the light guide member **113**.

Multiple light source placement portions **113*a*** are located in the light guide member **113**. The multiple light source placement portions **113*a*** are arranged in a matrix configuration in top-view. According to the present embodiment, as shown in FIG. **3**, each light source placement portion **113*a*** is a through-hole that extends through the light guide member **113** in the Z-direction. Alternatively, the light source placement portion **113*a*** may be a bottomed recess located at the lower surface of the light guide member **113**.

The light sources **114** are located in the light source placement portions **113*a***, respectively. Accordingly, as shown in FIG. **2**, multiple light sources **114** also are arranged in a matrix configuration. However, it is not always necessary for the light guide member **113** to be included in the planar light source **111**. For example, the planar light source **111** may not include a light guide member, and the multiple light sources **114** may simply be arranged in a matrix configuration on the substrate **112**. When no light guide member is included, the light source placement portion refers to a portion of the substrate **112** in which the light source **114** is located.

Each light source **114** may be a single light-emitting element or may include a light-emitting device in which, for example, a wavelength conversion member or the like is combined with a light-emitting element. According to the present embodiment as shown in FIG. **3**, each light source **114** includes a light-emitting element **114***a*, a wavelength conversion member **114***b*, a second light-modulating member **114***h*, and a third light-modulating member **114***i*.

The light-emitting element **114***a* is, for example, an LED (Light-Emitting Diode) and includes a semiconductor stacked body **114***c* and a pair of electrodes **114***d* and **114***e* that electrically connects the semiconductor stacked body **114***c* and the wiring of the substrate **112**. Through-holes are provided in portions of the light-reflective sheet **112***s* positioned directly under the electrodes **114***d* and **114***e*. Conductive members **112***m* that electrically connect the substrate **112** and the electrodes **114***d* and **114***e* are located in the through-holes.

The wavelength conversion member **114***b* includes a light-transmitting member **114***f* that covers an upper surface and side surfaces of the semiconductor stacked body **114***c*, and a wavelength conversion substance **114***g* that is located in the light-transmitting member **114***f* and converts the wavelength of the light emitted by the semiconductor stacked body **114***c* into a different wavelength. The wavelength conversion substance **114***g* is, for example, a phosphor.

According to the present embodiment, the light-emitting element **114***a* emits blue light. On the other hand, the wavelength conversion member **114***b* includes, for example, a phosphor that converts incident light into red light (hereinbelow, called a red phosphor) such as a CASN-based phosphor (e.g., $CaAlSiN_3$:Eu), a KSF-based phosphor (e.g., K_2SiF_6:Mn), a KSAF-based phosphor (e.g., $K_2(Si, Al)F_6$:Mn), or the like, a phosphor that converts incident light into green light (hereinbelow, called a green phosphor) such as a phosphor that has a perovskite structure (e.g., $CsPb(F, Cl, Br, I)_3$), a β-sialon-based phosphor (e.g., $(Si, Al)_3(O, N)_4$:Eu), a LAG-based phosphor (e.g., $Lu_3(Al, Ga)_5O_{12}$:Ce), etc. Thereby, the backlight **110** can emit white light, which is a combination of the blue light emitted by the light-emitting element **114***a* and the red light and the green light from the wavelength conversion member **114***b*. The wavelength conversion member **114***b* may be a light-transmitting member that does not include any phosphor; in such a case, for example, a similar white light can be obtained by providing a phosphor sheet that includes a red phosphor and a green phosphor on the planar light source.

The second light-modulating member **114***h* is located at an upper surface of the wavelength conversion member **114***b* and can modify the amount and/or the emission direction of the light emitted from the upper surface of the wavelength conversion member **114***b*. The third light-modulating member **114***i* is located at the lower surface of the light-emitting element **114***a* and the lower surface of the wavelength conversion member **114***b* so that the lower surfaces of the electrodes **114***d* and **114***e* are exposed. The third light-modulating member **114***i* can reflect the light oriented toward a lower surface of the wavelength conversion member **114***b* to the upper surface and side surfaces of the wavelength conversion member **114***b*. The second light-modulating member **114***h* and the third light-modulating member **114***i* each can include a light-transmitting resin, a light-diffusing agent included in the light-transmitting resin, etc. The light-transmitting resin is, for example, a silicone resin, an epoxy resin, or an acrylic resin. For example, particles of TiO_2, SiO_2, Nb_2O_5, $BaTiO_3$, Ta_2O_5, Zr_2O_3, Y_2O_3, Al_2O_3, ZnO, MgO, $BaSO_4$, glass, etc., are examples of the light-diffusing agent. The second light-modulating member **114***h* may also include a metal member such as, for example, Al, Ag, etc., so that the luminance directly above the light source **114** does not become too high.

The light-transmitting member **115** is located in the light source placement portion **113***a*. The light-transmitting member **115** covers the light source **114**. The first light-modulating member **116** is located on the light-transmitting member **115**. The first light-modulating member **116** can reflect a portion of the light incident from the light-transmitting member **115** and can transmit another portion of the light so that the luminance directly above the light source **114** does not become too high. The first light-modulating member **116** can include a member similar to the second light-modulating member **114***h* or the third light-modulating member **114***i*.

A partitioning trench **113***b* is provided in the light guide member **113** to surround the light source placement portions **113***a* in top-view. High noticeability of the halo phenomenon can be suppressed by the partitioning trench **113***b* reflecting a portion of the light from the light source **114**. The partitioning trench **113***b* extends in a lattice shape in the X-direction and the Y-direction. The partitioning trench **113***b* extends through the light guide member **113** in the Z-direction. Alternatively, the partitioning trench **112***b* may be a recess provided in the upper surface or the lower surface of the light guide member **113**. Also, the partitioning trench **112***b* may not be provided in the light guide member **113**.

The light-reflecting member **117** is located in the partitioning trench **113***b*. The high noticeability of the halo phenomenon can be further suppressed by the light-reflecting member **117** reflecting a portion of the light from the light source. For example, a light-transmitting resin that includes a light-diffusing agent can be used as the light-reflecting member **117**. For example, particles of TiO_2, SiO_2, Nb_2O_5, $BaTiO_3$, Ta_2O_5, Zr_2O_3, ZnO, Y_2O_3, Al_2O_3, MgO, $BaSO_4$, glass, etc., are examples of the light-diffusing agent. For example, a silicone resin, an epoxy resin, an acrylic resin, etc., are examples of the light-transmitting resin. For example, a metal member such as Al, Ag, etc., may be used as the light-reflecting member **117**. The light-reflecting member **117** covers a portion of side surfaces of the partitioning trench **113***b* in a layer shape. Alternatively, the light-reflecting member **117** may fill the entire interior of the partitioning trench **112***b*. Also, no light-reflecting member may be located in the partitioning trench **112***b*.

According to the present embodiment, light emission of the multiple light sources **114** is individually controllable by the driver **120** for the backlight. Here, "controllable light emission" means that switching between lit and unlit is possible, and the luminance in the lit state is adjustable. For example, the planar light source may have a structure in which the light emission is controllable for each light source, or may have a structure in which multiple light source groups are arranged in a matrix configuration, and the light emission is controllable for each light source group.

In the specification, subdivided regions of the planar light source each of which includes a light source or a light source group that are individually controllable are called "light-emitting regions". In other words, the light-emitting region means the minimum region of the backlight of which the luminance is controllable by local dimming. Accordingly, according to the present embodiment, similarly to the partitioning trench **113***b*, the regions of the planar light source **111** partitioned into a lattice shape correspond to light-emitting regions **110***s*.

Each light-emitting region **110***s* is rectangular. According to the present embodiment, one light source **114** is located in one light-emitting region **110***s*. Then, the luminances of the multiple light-emitting regions **110***s* are individually controlled by the driver **120** for the backlight individually controlling the light emission of the multiple light sources **114**. As described above, when the light emission is controlled for each of multiple light source groups, one light source group, i.e., multiple light sources, is located in one light-emitting region; and the multiple light sources are simultaneously lit or unlit.

The multiple light-emitting regions **110***s* are arranged in a matrix configuration in top-view. Hereinbelow, in the structure of a matrix configuration such as that of the multiple light-emitting regions **110***s*, the element group of the matrix of the light-emitting region **110***s*, etc., arranged in the X-direction is called a "row"; and the element group of the matrix of the light-emitting region **110***s*, etc., arranged in the Y-direction is called a "column". For example, as shown in FIG. **2**, the row that is positioned furthest in the +Y direction (the row positioned uppermost when viewed according to a direction of reference numerals) is referred to as the "first row"; and the row that is positioned furthest in the −Y direction (the row positioned lowermost when viewed according to the direction of reference numerals) is referred to as the "final row". Similarly, as shown in FIG. **2**, the column that is positioned furthest in the −X direction (the column positioned leftmost when viewed according to the direction of reference numerals) is referred to as the "first column"; and the column that is positioned furthest in the +X direction (the column positioned rightmost when viewed according to the direction of reference numerals) is referred to as the "final column". The multiple light-emitting regions **110***s* are arranged in N1 rows and M1 columns. Here, N1 and M1 each are any integer; an example is shown in FIG. **2** in which N1 is 8 and M1 is 16.

Although the partitioning trench **113***b* and the light-reflecting member **117** are included in the planar light source **111** as shown in FIG. **3**, the adjacent light-emitting regions **110***s* are not perfectly shielded. Therefore, light can propagate between the adjacent light-emitting regions **110***s*. Accordingly, the light that is emitted by the light source **114** in one light-emitting region **110***s* when the light source is lit may propagate to the adjacent light-emitting regions **110***s* at the periphery of the one light-emitting region **110***s*.

As shown in FIG. **1**, the driver **120** for the backlight is connected to the substrate **112** and the controller **150**. The driver **120** for the backlight includes a drive circuit that drives the multiple light sources **114**. The driver **120** for the backlight adjusts the luminances of the light-emitting regions **110***s* according to backlight control data SG**1** received from the controller **150**.

FIG. **4** illustrates a top view of the liquid crystal panel **130** of the image display device **100** according to the first embodiment.

The liquid crystal panel **130** is located on the backlight **110**. According to the present embodiment, the liquid crystal panel **130** is substantially rectangular in top-view. The liquid crystal panel **130** includes multiple pixels **130***p* arranged in a matrix configuration. In FIG. **4**, one region that is surrounded with a double dot-dash line corresponds to one pixel **130***p*.

The liquid crystal panel **130** according to the present embodiment can display a color image. To achieve this objective, one pixel **130***p* includes three subpixels **130***sp* such that, for example, the white light emitted from the backlight **110** is transmitted to a subpixel that is configured to transmit blue light, a subpixel that is configured to transmit green light, and a subpixel that is configured to transmit red light. The light transmittances of the subpixels **130***sp* are individually controllable by the driver **140** for the liquid crystal panel. The gradations of the subpixels **130***sp* are individually controlled thereby.

The multiple pixels **130***p* are arranged in N2 rows and M2 columns. Here, N2 and M2 each are any integer such that N2>N1 and M2>M1. The multiple pixels **130***p* are located in the light-emitting regions **110***s* in top-view. Although an example is shown in FIG. **4** demonstrates that four pixels **130***p* correspond to one light-emitting region **110***s*, the number of the pixels **130***p* that correspond to one light-emitting region **110***s* may be less than four or more than four.

As shown in FIG. **1**, the driver **140** for the liquid crystal panel is connected to the liquid crystal panel **130** and the controller **150**. The driver **140** for the liquid crystal panel includes a drive circuit of the liquid crystal panel **130**. The driver **140** for the liquid crystal panel adjusts gradations of the pixels **130***p* according to liquid crystal panel control data SG**2** received from the controller **150**.

FIG. **5** is a block diagram showing components of the image display device **100** according to the first embodiment.

According to the first embodiment, the controller **150** includes an input interface **151**, memory **152**, a processor **153** such as a CPU (central processing unit) or the like, and an output interface **154**. These components are connected to each other by a bus.

For example, the input interface **151** is connected to an external device **900** such as a tuner, a personal computer, a game machine, etc. The input interface **151** includes, for example, a connection terminal to the external device **900** such as a HDMI® (High-Definition Multimedia Interface) terminal, etc. The external device **900** inputs an input image IM to the controller **150** via the input interface **151**.

The memory **152** includes, for example, ROM (Read-Only Memory), RAM (Random-Access Memory), etc. The memory **152** stores various programs, various parameters, and various data for displaying an image in the liquid crystal panel.

By reading the programs stored in the memory **152**, the processor **153** processes the input image IM, determines setting values of luminances of the light-emitting regions **110***s* of the backlight **110** and setting values of the gradations of the pixels **130***p* of the liquid crystal panel **130**, and controls the backlight **110** and the liquid crystal panel **130** based on these setting values. Thereby, an image that corresponds to the input image IM is displayed on the liquid crystal panel **130**. The processor **153** includes a luminance data generator **153***a*, a luminance setting data generator **153***b*, a gradation setting data generator **153***c*, and a control unit **153***d*.

The output interface **154** is connected to the driver **120** for the backlight. Also, the output interface **154** includes, for example, a connection terminal of the driver **140** for the liquid crystal panel such as a HDMI® terminal, etc., and is connected to the driver **140** for the liquid crystal panel. The driver **120** for the backlight receives the backlight control data SG**1** via the output interface **154**. The driver **140** for the liquid crystal receives the liquid crystal panel control data SG**2** via the output interface **154**.

An image display method that uses the image display device **100** according to the present embodiment will be described hereinafter. Functions of the processor **153** as the luminance data generator **153***a*, the luminance setting data generator **153***b*, the gradation setting data generator **153***c*, and the control unit **153***d* also will be described.

FIG. **6** is a flowchart showing an image display method according to the first embodiment.

The image display method according to the first embodiment includes an acquisition process S1 of the input image

IM, a generation process S2 of luminance data D1, a generation process S3 of luminance setting data D2, a generation process S4 of gradation setting data D3, and a display process S5 of the image on the liquid crystal panel **130**. The processes will now be elaborated. A method of displaying an image corresponding to one input image IM on the liquid crystal panel **130** will be described. When the input images IM are sequentially input to the controller **150** and images that correspond to the input images IM are sequentially displayed on the liquid crystal panel **130**, the following process S1 to S5 are repeatedly performed.

First, the acquisition process S1 of the input image IM will be described.

As shown in FIG. **5**, the input interface **151** of the controller **150** receives the input image IM from the external device **900**. The received input image IM is stored in the memory **152**.

FIG. **7** is a schematic diagram showing an input image input to the controller **150** of the image display device **100** according to the first embodiment.

FIG. **8** is a schematic diagram showing a relationship among the pixels of the liquid crystal panel **130**, the light-emitting regions of the backlight **110**, and pixels of the input image the first embodiment.

The input image IM is data in which gradations are set for multiple pixels (may be referred to as "image pixels") IMp arranged in a matrix configuration. According to the first embodiment, the input image IM is a color image. To achieve this objective, a blue gradation Gb, a green gradation Gg, and a red gradation Gr are set for one pixel IMp. For example, the gradations Gb, Gg, and Gr are represented by numerals from 0 to 255.

For easier understanding of the following description, for example, the arrangement directions of the elements are represented using a xy orthogonal coordinate system for data in which elements such as the pixels IMp or the like are arranged in a matrix configuration as in the input image IM. The x-axis direction in the direction of the arrow is referred to as the "+x direction"; and the opposite direction is referred to as the "−x direction". Similarly, the y-axis direction in the direction of the arrow is referred to as the "+y direction"; and the opposite direction is referred to as the "−y direction". Also, hereinbelow, the element groups of the matrix that are arranged in the x-direction are called a "row"; and the element groups of the matrix that are arranged in the y-direction are called a "column". For example, as shown in FIG. **7**, the row that is positioned furthest in the +y direction (the row positioned uppermost when viewed according to a direction of reference numerals) is referred to as the "first row"; and the row that is positioned furthest in the −y direction (the row positioned lowermost when viewed according to the direction of reference numerals) is referred to as the "final row". Similarly, the column that is positioned furthest in the −x direction (the column positioned leftmost when viewed according to the direction of reference numerals) is referred to as the "first column"; and the column that is positioned furthest in the +x direction (the column positioned rightmost when viewed according to the direction of reference numerals) is referred to as the "final column".

For easier understanding of the following description, an example is described in which one pixel IMp of the input image IM corresponds to one pixel **130***p* of the liquid crystal panel **130** as shown in FIG. **8**. In other words, according to the present embodiment, the multiple pixels IMp are arranged in N2 rows and M2 columns. Then, multiple pixels IMp are included in an area IMs of the input image IM that corresponds to one light-emitting region **110***s* of the backlight **110**. However, the correspondence between the pixels of the input image and the pixels of the liquid crystal panel may not be one-to-one. In such a case, the processor **153** of the controller **150** performs the following processing after performing preprocessing of the input image so that the pixels of the input image and the pixels of the liquid crystal panel correspond one-to-one.

The generation process S2 of the luminance data D1 will now be described.

FIG. **9** is a schematic diagram showing a process of generating luminance data in the image display method according to the first embodiment.

The luminance data generator **153***a* generates the luminance data D1 including a luminance L converted from a maximum gradation Gmax of the gradations Gb, Gg, and Gr of the multiple pixels IMp with respect to each area IMs of the input image IM corresponding to one light-emitting regions **110***s*.

Specifically, first, the luminance data generator **153***a* determines an area IMs that corresponds to the light-emitting region **110***s* positioned at the ith row and the jth column. Then, the luminance data generator **153***a* uses the maximum value of the red gradation Gr, the green gradation Gg, or the blue gradation Gb of all pixels IMp included in the area IMs as the maximum gradation Gmax of the area IMs. Then, the luminance data generator **153***a* converts the maximum gradation Gmax into the luminance L. Then, the luminance data generator **153***a* uses the luminance L as a value of an element $e1(i, j)$ at the ith row and the jth column of the luminance data D1. Here, i is any integer from 1 to N1, and j is any integer from 1 to M1.

The luminance data generator **153***a* performs this processing for all of the areas IMs.

The luminance data D1 thus obtained is data of a matrix configuration that includes N1 rows and M1 columns. The value of the element $e1(i, j)$ of the luminance data D1 at the ith row and the jth column is the luminance L converted from the maximum gradation Gmax of the area IMs at the ith row and the jth column. In other words, the luminance data D1 is data of a matrix configuration in which the luminance L is set for each area IMs.

The luminance data generator **153***a* stores the luminance data D1 in the memory **152**.

FIG. **10** is a graph showing a luminance distribution when a light source in one light-emitting region is lit in the backlight of the image display device according to the first embodiment. In FIG. **10**, the horizontal axis is the position in the X-direction, and the vertical axis is the luminance.

In FIG. **10**, the light-emitting region **110***s* in which the light source **114** is lit is shown as ON, and the light-emitting regions **110***s* in which the light sources **114** are unlit are shown as OFF.

In the planar light source **111** according to the present embodiment, the adjacent light-emitting regions **110***s* are not perfectly shielded. Therefore, when the light source **114** in one light-emitting region **110***s* of the backlight **110** is lit, the light emitted from the light source **114** may propagate to neighboring light-emitting regions **110***s* at the periphery of the one light-emitting region **110***s*. For that reason, when the light source **114** in the one light-emitting region **110***s* is lit and the light sources **114** in the neighboring light-emitting regions **110***s* at the periphery of the one light-emitting region **110***s* are unlit, the luminances of the neighboring light-emitting regions **110***s* at the periphery are not perfectly zero. The leak of the light of the light source **114** in the brighter light-emitting regions **110***s* to the darker neighboring light-emitting regions **110***s* is highly noticeable as the luminance difference between the adjacent light-emitting regions **110***s* increases.

In a conventional image display device, the controller converts the luminance data D**1** into backlight control data as-is, and controls the driver for the backlight based on the converted backlight control data. Because the luminance data D**1** is determined solely according to the input image IM as is, the luminance difference between the adjacent light-emitting regions **110***s* may be large enough to cause high noticeability of a halo phenomenon depending on the input image IM. In contrast, the image display method according to the first embodiment can suppress the high noticeability of the halo phenomenon by performing the generation process S**3** of the luminance setting data D**2** that is described below.

The generation process S**3** of the luminance setting data D**2** will now be described.

FIG. **11** is a flowchart showing a process of correcting the luminance of one area of the luminance data in the image display method according to the first embodiment.

FIGS. **12**A to **15**B are schematic diagrams showing details of the process of generating the luminance setting data in the image display method according to the first embodiment.

As shown in FIGS. **13**A and **13**B, the luminance setting data generator **153***b* generates the luminance setting data D**2** including the setting values of the luminances of the light-emitting regions **110***s* of the backlight **110**. The luminance setting data D**2** is generated by correcting the luminance data D**1** to reduce a luminance difference ΔL by increasing the luminance L of the area IMs that has a lower luminance L among the adjacent areas IMs when a luminance difference ΔL between the adjacent areas IMs of the luminance data D**1** is greater than a threshold ΔLdet.

A specific example of the process of generating the luminance setting data D**2** will now be described.

For easier understanding of the following description, the area IMs that is at the ith row and the jth column of the luminance data D**1** also is called the "area IMs(i, j)". The luminance L of the area IMs at the ith row and the jth column also is called the "luminance L(i, j)".

First, as shown in FIGS. **11** and **12**A, the luminance setting data generator **153***b* refers to the luminance L(1, 1) of the area IMs(1, 1) at the first row and the first column of the luminance data D**1** and the luminances L(1, 2), L(2, 1), and L(2, 2) of the neighboring areas IMs(1, 2), IMs(2, 1), and IMs(2, 2) at the periphery of the area IMs(1, 1) (a process S**31**).

Then, as shown in FIGS. **11** and **12**A, the luminance setting data generator **153***b* determines a maximum value Lmax of the luminances L(1, 2), L(2, 1), and L(2, 2) of its neighboring areas IMs(1, 2), IMs(2, 1), and IMs(2, 2) at the periphery (a process S**32**). Here, the luminance L(2, 2) is the maximum value Lmax.

Next, as shown in FIGS. **11** and **13**A, the luminance setting data generator **153***b* calculates the luminance difference ΔL between the maximum value Lmax and the luminance L(1, 1) and determines whether or not the luminance difference ΔL is greater than the threshold ΔLdet (a process S**33**). In other words, the luminance setting data generator **153***b* determines whether or not the luminance difference ΔL is greater than the threshold ΔLdet.

The threshold ΔLdet is prestored in the memory **152**. According to the present embodiment, the threshold ΔLdet is a value that corresponds to a luminance difference such that a user that visually checks the image display device **100** is less likely to see the halo phenomenon. Although an example in the drawings shows that the threshold ΔLdet is 40, the specific numerical value of the threshold ΔLdet is not limited to 40.

When the luminance difference ΔL is determined to be greater than the threshold ΔLdet (the process S**33**: Yes), the luminance setting data generator **153***b* calculates a difference ΔLo between the luminance difference ΔL and the threshold ΔLdet as shown in FIGS. **11** and **13**A and calculates a value La(1, 1) by adding the difference ΔLo to the luminance L(1, 1) as shown in FIG. **13**B (the process S**34**). In other words, ΔLo=ΔL−ΔLdet, and La(1, 1)=L(1, 1)+ΔLo. Then, as shown in FIGS. **11** and **12**A, the luminance setting data generator **153***b* replaces the luminance L(1, 1) of the area IMs(1, 1) at the first row and the first column of the luminance data D**1** with this value La(1, 1).

When it is determined that the luminance difference ΔL is not more than the threshold ΔLdet (the process S**33**: No), the luminance setting data generator **153***b* does not correct the value of the luminance L(1, 1) as shown in FIG. **11** (the process S**36**). Hereinbelow, the luminance data D**1** on which the processing of the processes S**31** to S**35** or the processes S**31** to S**34** and S**36** is performed is called "corrected data D**1***a*".

Then, as shown in FIG. **12**B, the luminance setting data generator **153***b* refers to the luminance L(1, 2) of the area IMs(1, 2) at the first row and the second column of the corrected data D**1***a* and the luminances L(1, 1), L(1, 3), L(2, 1), L(2, 2), and L(2, 3) of its neighboring areas IMs(1, 1), IMs(1, 3), IMs(2, 1), IMs(2, 2), and IMs(2, 3) at the periphery of the area IMs(1, 2) (the process S**31**).

Next, as shown in FIGS. **11** and **12**B, the luminance setting data generator **153***b* calculates the maximum value Lmax of the luminances L(1, 1), L(1, 3), L(2, 1), L(2, 2), and L(2, 3) of the neighboring areas IMs(1, 1), IMs(1, 3), IMs(2, 1), IMs(2, 2), and IMs(2, 3) at the periphery (the process S**32**). Here, the luminances L(2, 2) and L(2, 3) are the maximum value Lmax.

Then, as shown in FIG. **11**, the luminance setting data generator **153***b* calculates the luminance difference ΔL between the maximum value Lmax and the luminance L(1, 1) and determines whether or not the luminance difference ΔL is greater than the threshold ΔLdet (the process S**33**).

When it is determined that the luminance difference ΔL is greater than the threshold ΔLdet (the process S**33**: Yes), the luminance setting data generator **153***b* calculates the difference ΔLo between the luminance difference ΔL and the threshold ΔLdet and calculates a value La(1, 2) by adding the difference ΔLo to the luminance L(1, 2) (the process S**34**). However, the value that is added to the luminance L(1, 2) is not limited to the difference ΔLo and may be any other value. Then, the luminance setting data generator **153***b* replaces the luminance L(1, 2) of the area IMs(1, 2) at the first row and the second column of the luminance data D**1** with this value La(1, 2).

When it is determined that the luminance difference ΔL is not more than the threshold ΔLdet (the process S**33**:No), the luminance setting data generator **153***b* does not correct the value of the luminance L(1, 2) as shown in FIG. **11** (the process S**36**).

In this manner, the luminance setting data generator **153***b* sequentially shifts the selected area IMs in the +x direction in the corrected data D**1***a*, and performs the processing of the processes S**31** to S**35** or the processes S**31** to S**33** and S**36** for each shift.

After the selected area IMs is shifted furthest in the +x direction in the corrected data D**1***a* as shown in FIG. **12**C, the luminance setting data generator **153***b* shifts the selected area IMs one row in the −y direction and furthest in the −x direction as shown in FIG. **14**A, and performs similar processing. Then, the luminance setting data generator **153***b* sequentially shifts the selected area IMs in the +x direction in the corrected data D**1***a* and performs similar processing for each shift.

In this manner, the luminance setting data generator **153***b* sequentially shifts the selected area IMs in the x-direction and/or the y-direction and performs the processing of the processes S**31** to S**35** or the processes S**31** to S**33** and S**36** for each shift. Then, finally, as shown in FIG. **14**B, the luminance setting data generator **153***b* performs the processing of the processes S**31** to S**35** or the processes S**31** to S**33** and S**36** for the area IMs(N1, M1) at the N1th row and the M1th column.

When only one cycle for all of the areas IMs from the area IMs(1, 1) to the area IMs(N1, M1) of the corrected data D**1***a* has been performed, the luminance difference ΔL may be still greater than the threshold ΔLdet for some of the areas IMs as in the area IMs(1, 1) and the area IMs(2, 2) of FIG. **15**A.

To address such an issue, according to the present embodiment, as shown in FIGS. **15**A and **15**B, the luminance setting data generator **153***b* again sequentially selects the areas IMs(1, 1) from the area IMs(1, 1) to the area IMs(N1, M1) and iteratively performs the processing of the processes S**31** to S**35** or the processes S**31** to S**33** and S**36** for each area IMs. In other words, two cycles are performed for all of the areas IMs from the area IMs(1, 1) to the area IMs(N1, M1). Thereby, the luminance difference ΔL between the adjacent areas IMs of the corrected data D**1***a* that is greater than the threshold ΔLdet can be suppressed.

The luminance setting data generator **153***b* uses the corrected data D**1***a* that is finally obtained as the luminance setting data D**2**.

The luminance setting data D**2** thus obtained is data of a matrix configuration of N1 rows and M1 columns. The value of each element e**2**(*i, j*) of the luminance setting data D**2** at the ith row and the jth column corresponds to the setting value of the luminance of the light-emitting region **110***s* positioned at the ith row and the jth column.

Then, the luminance setting data generator **153***b* stores the luminance setting data D**2** in the memory **152**.

As described above, the luminance setting data generator **153***b* generates the luminance setting data D**2** including the setting values of the luminances of the light-emitting regions **110***s* of the backlight **110** by correcting the luminance data D**1** to reduce the luminance difference ΔL by increasing the luminance L of the area IMs that has a lower luminance L among the adjacent areas IMs when the luminance difference ΔL is greater than the threshold ΔLdet. As a result, the luminance difference ΔL between the adjacent areas IMs of the luminance setting data D**2** can be less than the threshold ΔLdet. The halo phenomenon that is visible to the user of the image display device **100** can be suppressed thereby.

Although an example of the process of generating the luminance setting data D**2** is described above, the process of generating the luminance setting data is not limited to that described above. For example, the sequence of selecting the area IMs in the luminance data D**1** or the corrected data D**1***a* is not limited to the aforementioned sequence. Depending on the sequence of selecting the area IMs, the luminance difference ΔL may not become more than the threshold ΔLdet for all areas IMs for which the processing of the processes S**34** and S**35** has been performed once. In such a case, the processing of the processes S**31** to S**35** or the processes S**31** to S**33** and S**36** may be performed only once for each area IMs.

The generation process S**4** of the gradation setting data D**3** will now be described.

FIG. **16** is a schematic diagram showing a process of generating gradation setting data in the image display method according to the first embodiment.

The gradation setting data generator **153***c* generates gradation setting data D**3** including setting values of the gradations of the pixels **130***p* of the liquid crystal panel **130** based on the input image IM and the luminance setting data D**2**.

A specific example of the method for generating the gradation setting data D**3** will now be described.

According to the present embodiment, the memory **152** pre-stores luminance distribution data D**4** indicating luminance distribution in the XY plane when the light source **114** in one light-emitting region **110***s* is lit. Although the setting values of the luminances of the light-emitting regions **110***s* of the backlight **110** are determined in the process S**3**, actual luminance may be different depending on the position in the XY plane even in one light-emitting region **110***s* as shown in the luminance distribution data D**4** of FIG. **15**. Also, when the light source **114** in one light-emitting region **110***s* is lit, the light propagates to its neighboring light-emitting regions **110***s* at the periphery of the one light-emitting region **110***s* as described above.

To address such an issue, first, the gradation setting data generator **153***c* estimates a luminance value V(i, j) directly under the pixel **130***p* positioned at the ith row and the jth column of the liquid crystal panel **130** from the luminance setting data D**2** and the luminance distribution data D**4**.

Specifically, the gradation setting data generator **153***c* estimates a luminance value V**1**(*i, j*) of the luminance setting data D**2** directly under the pixel **130***p* when only the light source **114** in the light-emitting region **110***s* positioned directly under the pixel **130***p* is lit from the value of the element e**2** (the setting value of the luminance) corresponding to the light-emitting region **110***s* and the data D**4**. Furthermore, the gradation setting data generator **153***c* estimates a luminance value V**2**(*i, j*) of the luminance setting data D**2** directly under the pixel **130***p* when only the light sources **114** in the neighboring light-emitting regions **110***s* are lit from the values of the elements e**2** corresponding to the neighboring light-emitting regions **110***s* and the luminance distribution data D**4**. Then, the value of the sum of the luminance values V**1**(*i, j*) and V**2**(*i, j*) is estimated to be the luminance value V(i, j) directly under the pixel **130***p*. Thereby, the gradation setting data generator **153***c* can estimate the luminance value V(i, j) directly under the pixel **130***p* by including both the luminance distribution in the one light-emitting region **110***s* and the light leakage from the neighboring light-emitting regions **110***s*.

Then, the gradation setting data generator **153***c* inputs the estimated luminance value V(i, j) and the blue gradation Gb of the pixel Imp of the input image IM corresponding to the pixel **130***p*(i, j) into a conversion formula Ef. The conversion formula Ef is, for example, a conversion formula that converts the luminance into a gradation such as a gamma correction conversion formula, etc. The gradation setting data generator **153***c* uses an output value Efb of the conversion formula Ef generated by inputting the blue gradation Gb into the conversion formula Ef as the setting value of the blue gradation of the pixel **130***p*. Similar processing is performed also for the green gradation Gg; and an output value Efg of the conversion formula Ef obtained thereby is used as the setting value of the green gradation of the pixel **130***p*. The gradation setting data generator **153***c* performs similar processing also for the red gradation Gr; and an output value Efr of the conversion formula Ef obtained thereby is used as the setting value of the red gradation of the pixel **130***p*. The gradation setting data generator **153***c* uses the output values Efb, Efg, and Efr of the conversion formula Ef as the values of an element e**3**(*i, j*) at the ith row and the jth column of the gradation setting data D**3**.

The gradation setting data generator **153***c* performs this processing for each pixel **130***p* of the liquid crystal panel **130**. The gradation setting data D**3** is generated thereby.

The gradation setting data D**3** thus obtained is data of a matrix configuration of N2 rows and M2 columns. The three values of Efb, Efg, and Egr of the element e**3**(*i, j*) at the ith row and the jth column of the gradation setting data D**3** correspond respectively to the setting value of the blue gradation, the setting value of the green gradation, and the setting value of the red gradation of the pixel **130***p* positioned at the ith row and the jth column of the liquid crystal panel **130**.

The gradation setting data generator **153***c* stores the gradation setting data D**3** in the memory **152**.

Although an example of the process of generating the gradation setting data D**3** is described above, the process of generating the gradation setting data is not limited to that described above. For example, the luminance values may be input into the conversion formula after estimating the luminance values directly under all pixels of the liquid crystal panel.

The display process S**5** of the image will now be described.

The control unit **153***d* causes the liquid crystal panel **130** to display the image by controlling the backlight **110** based on the luminance setting data D**2** and by controlling the liquid crystal panel **130** based on the gradation setting data D**3**.

Specifically, as shown in FIG. **5**, the control unit **153***d* transmits the backlight control data SG**1** generated based on the luminance setting data D**2** to the driver **120** for the backlight via the output interface **154**. The backlight control data SG**1** is, for example, data of a PWM (Pulse Width Modulation) format, but is not particularly limited as long as the driver **120** for the backlight can operate based on the data. The driver **120** for the backlight controls the light emission of the light sources **114** based on the backlight control data SG**1**.

Also, the control unit **153***d* transmits the gradation setting data D**3**, which is the liquid crystal panel control data SG**2**, to the driver **140** for the liquid crystal panel via the output interface **154**. Alternatively, the liquid crystal panel control data SG**2** may be data converted from the gradation setting data D**3** into a format that enables the driving of the driver **140** for the liquid crystal panel. The driver **140** for the liquid crystal panel controls the pixels **130***p*, and more specifically, light transmittances for the light of the subpixels **130***sp* based on the liquid crystal panel control data SG**2**.

The timing of converting the luminance setting data D**2** into the backlight control data SG**1** is not particularly limited as long as the timing is in or after the process S**3**. When converting the gradation setting data D**3** into the liquid crystal panel control data SG**2**, the timing of the conversion is not particularly limited as long as the timing is in or after the process S**4**.

Effects of the first embodiment will now be described.

The image display method according to the first embodiment includes the process S**2** of generating the luminance data D**1**, the process S**3** of generating the luminance setting data D**2**, the process S**4** of generating the gradation setting data D**3**, and the process S**5** of displaying the image in the liquid crystal panel **130**.

The backlight **110** includes the multiple light-emitting regions **110***s* arranged in a matrix configuration. The liquid crystal panel **130** includes the multiple pixels **130***p*. The input image IM is input to the controller **150** of the image display device **100**.

In the process S**2**, the luminance data D**1** including the luminance L converted from the maximum gradation Gmax of an area Ims of the input image IM for each of the areas Inns corresponding to the light-emitting regions **110***s* of the backlight **110** is generated.

In the process S**3**, the luminance setting data D**2** including the setting values of the luminances of the light-emitting regions **110***s* of the backlight **110** is generated by correcting the luminance data D**1** to reduce the luminance difference ΔL by increasing the luminance L of the area Inns that has a lower luminance L when the luminance difference ΔL is greater than the threshold ΔLdet.

In the process S**4**, the gradation setting data D**3** including the setting values of the gradations of the pixels **130***p* of the liquid crystal panel **130** is generated based on the luminance setting data D**2** and the input image IM.

In the process S**5**, the image is displayed on the liquid crystal panel **130** by controlling the backlight **110** based on the luminance setting data D**2** and by controlling the liquid crystal panel **130** based on the gradation setting data D**3**.

In such a manner, in the image display method according to the first embodiment, the luminance data D**1** is corrected to reduce the luminance difference ΔL between the adjacent areas Inns. Therefore, compared to the case where the backlight **110** is controlled based on the luminance data D**1** as is, according to the first embodiment, the difference between the setting values of the luminances of the adjacent light-emitting regions **110***s* of the backlight can be reduced. As a result, the halo phenomenon can be suppressed.

If, for example, a correction is performed to reduce the luminance L of the area Ims that has a higher luminance L among the adjacent areas Ims to reduce the luminance difference ΔL, the backlight **110** becomes darker. When the backlight **110** becomes darker, the gradation difference of the input image IM may no longer be sufficiently represented on the liquid crystal panel **130**. In contrast, according to the first embodiment, a correction is performed to increase the luminance L of the area Ims that has a lower luminance L among the adjacent areas Ims to reduce the luminance difference ΔL. As a result, the backlight **110** becomes brighter. Therefore, the liquid crystal panel **130** easily expresses the gradation difference of the input image IM compared to the case where the correction is performed to reduce the luminance L of the area Inns that has a higher luminance L among the adjacent areas Inns to reduce the luminance difference ΔL.

In the process S**3** of generating the luminance setting data D**2**, the difference ΔLo between the threshold ΔLdet and the luminance difference ΔL is added to the luminance L of the area Inns that has a lower luminance L when the luminance difference ΔL between the adjacent areas Inns of the luminance data D**1** is greater than the threshold ΔLdet. Thereby, the difference between the setting values of the luminances of the adjacent light-emitting regions **110***s* can be not more than the threshold ΔLdet. As a result, the halo phenomenon can be suppressed.

The image display device **100** according to the first embodiment includes: the backlight **110** including the planar light source **111** that includes the multiple light-emitting regions **110***s* arranged in a matrix configuration and includes the light sources **114** located in the multiple light-emitting regions **110***s*; the liquid crystal panel **130** that is positioned on the backlight **110** and includes the multiple pixels **130***p*; and the controller **150** controlling the backlight **110** and the liquid crystal panel **130**. The controller **150** includes the luminance data generator **153***a*, the luminance setting data generator **153***b*, the gradation setting data generator **153***c*, and the control unit **153***d*.

The luminance data generator **153***a* generates the luminance data D**1** by converting the maximum gradation Gmax of an area Ims of the input image IM into the luminance L for each of the areas Ims corresponding to the light-emitting regions **110***s* of the backlight **110**.

The luminance setting data generator **153***b* generates the luminance setting data D**2** including the setting values of the luminances of the light-emitting regions **110***s* of the backlight **110** by correcting the luminance data D1 to reduce the luminance difference ΔL by increasing the luminance L of the area Ims that has a lower luminance L among the adjacent areas Ims when the luminance difference ΔL is greater than the threshold ΔLdet.

The gradation setting data generator **153***c* generates the gradation setting data D**3** including the setting values of the gradations of the pixels **130***p* of the liquid crystal panel **130** based on the luminance setting data D2 and the input image IM.

The control unit **153***d* displays the image on the liquid crystal panel **130** by controlling the backlight **110** based on the luminance setting data D**2** and by controlling the liquid crystal panel **130** based on the gradation setting data D**3**.

As a result, compared to the case where the backlight **110** is controlled based on the luminance data D1 as is, according to the first embodiment, the difference between the setting values of the luminances of the adjacent light-emitting regions **110***s* of the backlight can be reduced. As a result, the halo phenomenon can be suppressed.

Second Embodiment

A second embodiment will now be described.

FIG. **17** is a flowchart showing a process of generating luminance setting data in an image display method according to the second embodiment.

The image display method according to the second embodiment differs from the image display method according to the first embodiment in that a luminance setting data D**22** is generated by applying a spatial filter F to the corrected data D**1***a*.

As a general rule in the following description, only the differences from the first embodiment are described. Other than the aspects described below, the second embodiment is similar to the first embodiment.

According to the second embodiment, the process S3 of generating the luminance setting data includes a sub-process S3*a* of generating the corrected data D1*a* in which the luminance difference ΔL is reduced by increasing the luminance L of the area IMs that has a lower luminance L when the luminance difference ΔL is greater than the threshold ΔLdet, and a sub-process S3*b* of generating the luminance setting data D22 by reducing the luminance difference ΔL between the adjacent areas IMs of the corrected data D1*a* by applying the spatial filter F to the corrected data D1*a*. The sub-process S3*a* is the same as the process S3 described in the first embodiment, and a detailed description is therefore omitted.

The sub-process S3*b* will now be elaborated.

FIGS. **18** to **20** are schematic diagrams showing details of the process of generating the luminance setting data in the image display method according to the second embodiment.

As shown in FIG. **20**, the luminance setting data generator **153***b* applies the spatial filter F to the corrected data D1*a* that is finally obtained by performing the processing of the processes S**31** to S**35** or the processes S**31** to S**33** and S**36** as described in the first embodiment for each area IMs of the luminance data D**1**.

The spatial filter F is prestored in the memory **152**. According to the second embodiment, the spatial filter F includes multiple weighting factors Fw arranged in a matrix configuration. Although in an example in the second embodiment the spatial filter F is a matrix of three rows and three columns, the number of rows and the number of columns of the spatial filter F are not limited to the aforementioned numbers. Hereinbelow, the weighting factor Fw at the ith row and the jth column also is called the weighting factor Fw(i, j). Here, i and j each are any integer from 1 to 3.

It is favorable for the value of the weighting factor Fw(2, 2) at the center of the spatial filter F to be greater than the values of the other weighting factors Fw. A Gaussian filter is shown as an example of the spatial filter F in FIGS. **19** and **20** in which the value of the weighting factor Fw(2, 2) at the center is greater than the values of the other weighting factors Fw. However, the values of the weighting factors of the spatial filter are not particularly limited as long as the luminance difference between the adjacent areas can be reduced. For example, the spatial filter may be an averaging filter or a median filter. According to the embodiment, the sum total of the weighting factors Fw is 1.

A specific example of the sub-process of applying the spatial filter F will now be described.

First, as shown in FIG. **18**, the luminance setting data generator **153***b* adds elements at the periphery of the corrected data D**1***a* so that the values of the elements are equal to the values of the adjacent elements. Thereby, the corrected data D**1***a* is enlarged, and the number of rows of the luminance setting data D**2** finally obtained can match the number of rows of the light-emitting regions **110***s*. Also, the number of columns of the luminance setting data D2 finally obtained can match the number of columns of the light-emitting regions **110***s*. The corrected data may be enlarged by adding elements of which the values are zero (0) to the periphery. In other words, zero padding of the corrected data may be performed.

Hereinbelow, the enlarged luminance data D1 is called the "post-enlargement corrected data Diaz". Elements of the post-enlargement corrected data Diaz are called "elements e1*a*". The element e1*a* is one of an element of which the value is the luminance L calculated in the process S**2**, an element of which the value is the value La in which the luminance L calculated in the sub-process S3*a* is corrected, or an added element of which the value is equal to the values of the adjacent elements.

Then, as shown in FIG. **19**, the luminance setting data generator **153***b* extracts a region Af that is positioned furthest in the −x direction and furthest in the +y direction in the post-enlargement corrected data Diaz and has the same size as the spatial filter F. Hereinbelow, the element e1*a* at the ith row and the jth column in this region Af is also called the element e**1***a*(i, j).

Continuing, the luminance setting data generator **153***b* calculates the product of e**1***a*(i, j)×Fw(i, j) in which the element e**1***a*(i, j) at the ith row and the jth column in this region Af and the weighting factor Fw(i, j) at the ith row and the jth column of the spatial filter F are multiplied. The luminance setting data generator **153***b* performs the calculation of the product of e**1***a*(i, j)×Wf(i, j) for all elements e**1***a*(i, j) included in this region Af.

Then, the luminance setting data generator **153***b* calculates a sum Sf(1, 1) of the products of e**1***a*(i, j)×Fw(i, j) calculated for one region Af. Thus, for two matrixes such as the region Af and the spatial filter F, the products of the elements at the same positions (coordinates) are calculated, and the sum of the calculated products is called the "multiply-add operation".

Next, the luminance setting data generator **153***b* uses the sum Sf(1, 1) as the value of an element e**22**(1, 1) at the first row and the first column of the luminance setting data D**22**.

Then, likewise, the luminance setting data generator **153***b* sequentially shifts the region Af in the +x direction and performs the multiply-add operation for each shift. When the region Af is positioned furthest in the +x direction, the luminance setting data generator **153***b* shifts the region Af one row in the −y direction and furthest in the −x direction, and performs the multiply-add operation. Then, the luminance setting data generator **153***b* again shifts the region Af one column at a time in the +x direction and performs the multiply-add operation for each shift. Thus, the luminance setting data generator **153***b* sequentially shifts the region Af in the x-direction and/or the y-direction and performs the multiply-add operation for each shift.

Finally, as shown in FIG. **20**, the region Af subjected to the multiply-add operation is positioned furthest in the +x direction and furthest in the −y-direction. Then, the luminance setting data generator **153***b* performs the multiply-add operation of the element e**1***a*(i, j) included in this region Af and the weighting factor Fw(i, j) of the spatial filter F. The sum Sf(N1, M1) is calculated thereby. Then, the luminance setting data generator **153***b* uses the sum Sf(N1, M1) as the value of an element e**22**(N1, M1) at the final row and the final column of the luminance setting data D**22**.

The luminance setting data D**22** thus obtained is data of a matrix configuration of N1 rows and M1 columns. The value of each element e**22**(*n, m*) of the luminance setting data D**22** at the nth row and the mth column corresponds to the setting value of the luminance of the light-emitting region **110***s* positioned at the nth row and the mth column.

Then, the luminance setting data generator **153***b* stores the luminance setting data D**22** in the memory **152**.

In the image display method according to the second embodiment as described above, the process S**3** of generating the luminance setting data D**22** includes the sub-process S**3***a* of generating the corrected data D**1***a* by reducing the luminance difference ΔL by increasing the luminance L of the area IMs that has a lower luminance L when the luminance difference ΔL is greater than the threshold ΔLdet, and the sub-process S**3***b* of generating the luminance setting data D**22** by reducing the luminance difference ΔL between the adjacent areas IMs of the corrected data D**1***a* by applying the spatial filter F to the corrected data D**1***a*.

In such a manner, by applying the spatial filter F to the corrected data D**1***a*, compared to the case where the backlight **110** is controlled based on the luminance data D**1** as is, the difference between the setting values of the luminances of the adjacent light-emitting regions **110***s* of the backlight can be reduced even further. As a result, the halo phenomenon can be suppressed.

For example, the invention can be utilized in the display of a device such as a television, a personal computer, a game machine, etc.

What is claimed is:

1. An image display method comprising:

generating luminance data that comprises data elements of a plurality of rows and columns of a backlight matrix, based on a maximum gradation value among gradation values of image pixels of an input image that correspond to the light-emitting region, wherein each data element indicates a luminance value for a corresponding one of a plurality of light-emitting regions of the backlight;

generating luminance setting data from the luminance data;

generating gradation setting data that sets a gradation value of each of a plurality of pixels of a liquid crystal panel coupled to the backlight for the input image, based on the input image and the luminance setting data; and

controlling the backlight to operate based on the luminance setting data, and the liquid crystal panel to operate based on the gradation setting data to display an image corresponding to the input image, wherein

said generating luminance setting data comprises:

performing correction of the luminance data, wherein with respect to each of the data elements of the luminance data of which luminance value is less than a maximum luminance value among luminance values of neighboring rows and columns thereof by more than a threshold amount, the luminance value is increased by a certain value such that a difference between the luminance value and the maximum luminance value is decreased to be within the threshold amount, to generate collected luminance data;

enlarging the collected luminance data by expanding rows and columns around rows and columns of the collected luminance data and setting luminance values of the expanded rows and columns, to generate enlarged luminance data; and

applying a spatial filter to each of different subsets of the enlarged luminance data, such that, with respect to each of data elements of the enlarged luminance data corresponding to the light-emitting regions, a difference of a luminance value thereof from luminance values of neighboring rows and columns decreases, to generate the luminance setting data, wherein

the spatial filter comprises a matrix of weighting factors, and

in said applying the spatial filter to each of different subsets of the enlarged luminance data, a sum of a multiplication of each luminance value in the subset with a corresponding one of the weighting factors is calculated, and the calculated sum is set as a luminance setting value for a corresponding light-emitting region of the backlight in the luminance setting data.

2. The image display method according to claim **1**, wherein the correction of the luminance data comprises:

performing iterative operations, each of the operations being carried out such that, with respect to each of the data elements of the luminance data of which luminance value is less than a maximum luminance value among luminance values of neighboring rows and columns thereof by more than the threshold amount, the luminance value is increased such that the difference between the luminance value and the maximum luminance value is decreased to be within the threshold amount.

3. The image display method according to claim 1, wherein the correction of the luminance data comprises:

with respect to each of the data elements of the luminance data,

determining whether the luminance value thereof is less than the maximum luminance value among the luminance values of the neighboring rows and columns thereof by more than the threshold amount; and

upon determining that the luminance value is less than the maximum luminance value by more than the threshold amount, increasing the luminance value by the certain value.

4. The image display method according to claim 3, wherein the certain value is a difference of the luminance value and the maximum luminance value, minus a predetermined value.

5. The image display method according to claim 3, wherein during the correction of the luminance data, with respect to each of the data elements of the luminance data, the luminance value thereof is maintained when the luminance value is determined to be within the predetermined range.

6. The image display method according to claim 1, wherein the neighboring rows and columns of each of the data elements of the luminance data are within one row and one column from the data element.

7. The image display method according to claim 1, wherein the spatial filter includes a Gaussian filter.

8. The image display method according to claim 1, wherein a sum of weighting factors of the spatial filter is one.

9. The image display method according to claim 1, wherein said generating gradation setting data comprises:

generating luminance estimation data that indicates an estimated luminance value of backlight for the input image with respect to each of the pixels of the liquid crystal panel based on the luminance setting data and luminance distribution data indicating a luminance distribution in each of the light-emitting regions of the backlight panel; and

performing correction of gradation values of image pixels indicated by the input image using the luminance estimation data to generate the gradation setting data.

10. The image display method according to claim 1, wherein each of the light-emitting regions of the backlight panel corresponds to a plurality of pixels of the liquid crystal panel.

11. The image display method according to claim 1, wherein each of the light-emitting regions of the backlight corresponds to a single light-emitting element.

12. The image display method according to claim 1, wherein

the expanded rows and columns are one row added to each of end rows of the collected luminance data and one column added to each of end columns of the collected luminance data, and

the matrix of the spatial filter is a three-by-three matrix.

13. A display comprising:

a backlight including a plurality of light-emitting regions that are configured in a matrix form and independently operable;

a liquid crystal panel coupled to the backlight panel and including a plurality of pixels; and

a controller configured to:

generate luminance data that comprises data elements of a plurality of rows and columns of the backlight matrix, based on a maximum gradation value among gradation values of image pixels of an input image that correspond to the light-emitting region, wherein each data element indicates a luminance value for a corresponding one of the light-emitting regions of the backlight;

generate luminance setting data from the luminance data;

generating gradation setting data that sets a gradation value of each of the pixels of the liquid crystal panel for the input image, based on the input image and the luminance setting data; and

control the backlight to operate based on the luminance setting data, and the liquid crystal panel to operate based on the gradation setting data to display an image corresponding to the input image,

wherein in generating the luminance setting data, the controller is configured to:

perform correction of the luminance data, wherein with respect to each of the data elements of the luminance data of which luminance value is less than a maximum luminance value among luminance values of neighboring rows and columns thereof by more than a threshold amount, the luminance value is increased by a certain value such that a difference between the luminance value and the maximum luminance value is decreased to be within the threshold amount, to generate collected luminance data;

enlarge the collected luminance data by expanding rows and columns around rows and columns of the collected luminance data and setting luminance values of the expanded rows and columns, to generate enlarged luminance data; and

apply a spatial filter to each of different subsets of the enlarged luminance data, such that, with respect to each of data elements of the enlarged luminance data corresponding to the light-emitting regions, a difference of a luminance value thereof from luminance values of neighboring rows and columns decreases, to generate the luminance setting data, wherein

the spatial filter comprises a matrix of weighting factors, and

in applying the spatial filter to each of different subsets of the enlarged luminance data, the controller is configured to calculate a sum of a multiplication of each luminance value in the subset with a corresponding one of the weighting factors, and set the calculated sum as a luminance setting value for a corresponding light-emitting region of the backlight in the luminance setting data.

14. The display according to claim 13, wherein the controller is configured to perform, during the correction of the luminance data, iterative operations, each of the operations being carried out such that, with respect to each of the data elements of the luminance data of which luminance value is less than a maximum luminance value among luminance values of neighboring rows and columns thereof by more than the threshold amount, the luminance value is increased such that the difference between the luminance value and the maximum luminance value is decreased to be within the threshold amount.

15. The display according to claim 13, wherein the controller is configured to, during the correction of the luminance data:

with respect to each of the data elements of the luminance data,

determine whether the luminance value thereof is less than the maximum luminance value among the luminance values of the neighboring rows and columns thereof by more than the threshold amount; and

upon determining that luminance value is less than the maximum luminance value by more than the threshold amount, increase the luminance value by the certain value.

16. The display according to claim **15**, wherein the certain value is a difference of the luminance value and the maximum luminance value, minus a predetermined value.

17. The display according to claim **15**, wherein during the correction of the luminance data, with respect to each of the data elements of the luminance data, the luminance value thereof is maintained when the luminance value is determined to be within the predetermined range.

18. The display according to claim **13**, wherein the neighboring rows and columns of each of the data elements of the luminance data are within one row and one column from the data element.

19. The display according to claim **13**, wherein the spatial filter includes a Gaussian filter.

20. The display according to claim **13**, wherein

the expanded rows and columns are one row added to each of end rows of the collected luminance data and one column added to each of end columns of the collected luminance data, and the matrix of the spatial filter is a three-by-three matrix.

* * * * *